



Residual-augmented IVX predictive regression

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ABSTRACT

Bias correction in predictive regressions is known to reduce the empirical size problems of OLS-based predictability tests with persistent predictors. This paper shows that bias correction is also achieved in the context of the extended instrumental variable (IVX) predictability testing framework introduced by Kostakis et al. (2015). To be specific, new IVX-based statistics subject to a bias correction analogous to that proposed by Amihud and Hurvich (2004) are introduced. Four important contributions are provided: first, we characterize the effects that bias-reduction adjustments have on the asymptotic distributions of the IVX test statistics in a general context allowing for short-run dynamics and heterogeneity; second, we discuss the validity of the procedure when predictors are stationary as well as near-integrated; third, we conduct an exhaustive Monte Carlo analysis to investigate the small in- and out-of-sample properties of the test procedures and their sensitivity to distinctive features that characterize predictive regressions in practice, such as strong persistence, endogeneity, and non-Gaussian innovations; and fourth, we provide an analysis of real estate return and rent growth predictability in 19 OECD countries.

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1. Introduction

Predictive regressions are widely used in economics and finance; see, e.g., Campbell (2008) and Phillips (2015) for surveys. The variable of interest (typically stock returns) is regressed on lagged values of a predictor and the existence of predictability assessed through the statistical significance of the resultant estimate of the corresponding slope parameter. However, two important features of putative predictors need to be taken into consideration, namely: (i) the possibly highly persistent autoregressive dynamics of predictors; and (ii) the strong contemporaneous correlation of the predictors' innovations with those of the dependent variable. These features raise serious problems of endogeneity which can lead to sizably biased estimates in finite samples and over-rejections of the null hypothesis of no predictability (Stambaugh, 1986; Mankiw and Shapiro, 1986). Moreover, biased estimates of slope parameters typically lead to lower-quality forecasts.

The usual asymptotic approximation employing the (standard) normal distribution performs particularly bad when predictors are persistent, even though the largest autoregressive roots of typical predictor candidates are usually smaller than one — reason for which near-integrated asymptotics has been favored as an alternative framework for inference (Elliott and Stock, 1994; Campbell and Yogo, 2006). In the context of near-integrated regressors, the limiting distribution of the slope parameter estimator is not centered at zero, and this bias depends on the mean reversion parameter of the near-integrated regressor.

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The uncertainty of the exact degree of persistence of a given predictor, and thus which critical values to use for a predictability test, pose an important challenge to empirical work. Moreover, standard estimation or pretests also fail in this context (Cavanagh et al., 1995). Similarly, regression misspecification tests are difficult to conduct; e.g. Georgiev et al. (2019) propose for this reason a fixed-regressor wild bootstrap implementation of a residual stationarity test. These difficulties have led to the proposal of a number of alternative approaches; see for instance, Campbell and Yogo (2006), Jansson and Moreira (2006), Maynard and Shimotsu (2009), Camponovo (2015), Breitung and Demetrescu (2015) and references therein. The recently proposed extended instrumental variable estimation approach of Kostakis et al. (2015) [denoted as IVX], motivated by the work of Magdalinos and Phillips (2009), is becoming increasingly popular in predictive regressions, especially because the relevant t -statistic exhibits the same limiting distribution in both stationary and near-integrated setups and is in this sense invariant to persistence; see, e.g., Gonzalo and Pitarakis (2012), Phillips and Lee (2013), Lee (2016) or Demetrescu et al. (2020b). The reasoning behind the approach consists in the generation of an instrumental variable whose persistence can be controlled, and this is achieved by suitably filtering the actual predictor.

To some extent, all methods considered in the predictive regression literature lose some power by having to robustify against unknown persistence; however, as illustrated by Kostakis et al. (2015) the IVX methodology offers a good balance between improved size control and power loss. Given however that the signal-to-noise ratio in predictive regressions is quite low, one should strive to further improve this balance. For instance, Demetrescu (2014b) uses a simple variable addition scheme to improve the convergence rates of IVX estimators (and thus the local power of the corresponding t -tests) when the instrument used is close to stationarity. Yet, for instrument choices closer to near-integration a different approach is required to improve the finite sample power of IVX-based tests without giving up size control.

To this end, we take a closer look at the class of reduced-bias techniques proposed by Amihud and Hurvich (2004) and extended by Amihud et al. (2009, 2010); see, *inter alia*, Bali (2008), Chun (2009), Avramov et al. (2013) and Johannes et al. (2014) for recent empirical applications building on this approach. When compared to other available procedures, the distinctive characteristic of these techniques is that they estimate the predictive slope coefficient and its standard error in a suitably *augmented* predictive regression, so that the bias is reduced to a minimum and the size properties of OLS-based predictability tests become stable. Moreover, bias reduction potentially benefits forecasts constructed using the fitted predictive regressions, since, in the typical predictive regression case, bias in the estimated slope coefficients is leveraged by multiplication with a predictor of uncertain persistence. Finally, we argue that bias reduction also has the potential to contribute to power improvements.

This paper discusses in more detail the large-sample behavior of a class of modified IVX statistics. Concretely, we conduct IVX based estimation and inference in an augmented predictive regression context analogous to that of Amihud and Hurvich (2004), and consider both stationary and near integrated predictors. Our main objectives are fourfold: (i) to characterize the effects that our adjustments have on the asymptotic distribution of the IVX-statistics in a general context; (ii) to establish the validity of the procedure when predictors are stationary as well as near-integrated; (iii) to provide an exhaustive Monte Carlo analysis to investigate the small in- and out-of-sample properties of forecast and test procedures under distinctive conditions that characterize predictive regressions in practice, such as strong persistence, endogeneity, and non-Gaussian innovations, and to contrast them to the properties of the IVX approach proposed by Kostakis et al. (2015); and (iv) to conduct an analysis of real estate returns and rent growth predictability for 19 OECD countries.

In summary, we establish that the reduced-bias techniques analyzed in this paper are useful in the reduction of the estimation bias in predictive regressions; are useful to reduce the forecast risk when augmented regressions are used for one-step ahead forecasts, particularly with persistent predictors; and are useful to stabilize the empirical size of predictability tests and, to some extent, to improve power in finite samples. For the analyzed data set, we find that the rent-to-price ratio (in nominal terms) is a useful predictor for both future housing returns and rent growth for Australia, Germany, Portugal, Spain and Japan.

The remainder of the paper is organized as follows: Section 2 describes the background of our analysis. Section 3 presents the large-sample theory under empirically relevant assumptions, including for instance time-varying unconditional variances. Section 4 discusses the finite sample performance, both in- and out-of-sample, of several procedures used to test for predictability. Section 5 presents the empirical analysis of the predictive power of rent-to-price ratios to predict returns and rent growth, and Section 6 summarizes and concludes the paper. A technical appendix collects the proofs of the main theoretical statements put forward in the paper.

2. Predictive regression framework and tests

2.1. The model

The typical predictive model characterizes the joint dynamics of a stochastic process, $\{y_t\}_{t=2}^T$, and its posited predictor, $\{x_t\}_{t=1}^{T-1}$, in a two-equation linear system as,

$$y_t = \alpha + \beta x_{t-1} + u_t, \quad t = 2, \dots, T \quad (1)$$

$$x_t = \mu_x + e_t, \quad e_t = \rho e_{t-1} + v_t; \quad (2)$$

see, *inter alia*, Stambaugh (1999), Amihud and Hurvich (2004), and Campbell and Yogo (2006). The model, which allows for empirically relevant features frequently found in economic and financial time series, is formalized next in Assumptions 1–4.

Importantly, we consider two different theoretical frameworks that critically determine the stochastic properties of the predictive variable. On the one hand, we consider stationary predictors, characterized by a fixed coefficient $|\rho| < 1$ in (2), and on the other, we allow for near-integration by considering $\rho := 1 - cT^{-1}$, with $c \geq 0$ and fixed. The main objective of this setting is to acknowledge the uncertainty that researchers face regarding the stochastic properties of the predictor, i.e., whether it is stable or near-integrated when $\hat{\rho}$ is close to, but strictly less than unity in finite samples. This setting includes the extreme case of a unit-root when the local to unity parameter equals zero ($c = 0$). We also allow for heterogeneity in the form of time-varying variances, different shapes of the distributions, and even changes in the persistence of the regressor.

Assumption 1. The data are generated according to (1)–(2) with initial condition e_1 bounded in probability.

Assumption 2. The errors u_t and v_t in (1) and (2), respectively, are given as

$$\begin{aligned} v_t &= a_1 v_{t-1} + \cdots + a_{p-1} v_{t-p+1} + v_t \\ u_t &= \gamma v_t + \varepsilon_t, \quad t \in \mathbb{Z}, \end{aligned}$$

where the innovations $(\varepsilon_t, v_t)'$ are contemporaneously orthogonal white noise as indicated below.

Assumption 3. Let

$$\begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix} := \begin{pmatrix} \sigma_{\varepsilon t} \xi_{\varepsilon t} \\ \sigma_{v t} \xi_{v t} \end{pmatrix}$$

where $(\xi_{\varepsilon t}, \xi_{v t})'$ is a heterogeneous independent sequence with unity covariance matrix and, for some $\delta > 0$, with uniformly bounded moments $E(|\xi_{\varepsilon t}^{4+\delta}|)$ and $E(|\xi_{v t}^{4+\delta}|)$. Furthermore, let $\sigma_{\varepsilon t} := \sigma_{\varepsilon}(t/T)$ and $\sigma_{v t} := \sigma_v(t/T)$, where $\sigma(\cdot)$ are piecewise Lipschitz continuous functions on $(-\infty, 1]$, bounded and bounded away from zero.

Assumption 4. The autoregressive parameter ρ in (2) is either (i) fixed when $|\rho| < 1$, or (ii) time-varying near unity, $\rho := 1 - c_t/T$ with $c_t := c(t/T)$ and $c(\cdot)$ is piecewise Lipschitz on $[0, 1]$.

The $\text{AR}(p-1)$ structure of v_t in Assumption 2 is taken as an approximation to more general data generating processes [DGP]s. In theory, this would require letting $p \rightarrow \infty$ at suitable rates as $T \rightarrow \infty$; dealing with such asymptotics is beyond the scope of this paper, but relevant results can be found, for instance, in Chang and Park (2002).

Assumption 3 acknowledges that time series may exhibit permanent volatility changes, which is an important stylized feature of many financial series; see, *inter alia*, Guidolin and Timmermann (2006), Teräsvirta and Zhao (2011) and Amado and Teräsvirta (2013). Since, such forms of nonstationarity typically invalidate the usual standard errors,¹ we resort to heteroskedasticity robust [HC] standard errors (also known as Eicker–White standard errors) to account for this feature. The use of Eicker–White standard errors is also recommended by Kostakis et al. (2015) to deal with conditional heteroskedasticity.² Kostakis et al. (2015) discuss however a specific GARCH case imposing strict stationarity, whereas we relax the i.i.d. assumption by replacing stationarity with smoothly varying volatility. (The case with both conditional and unconditional heteroskedasticity of generic form in the innovations has only recently been dealt with by Demetrescu et al., 2020a.)

Finally, Assumption 4 characterizes the persistence of the putative predictor. The flexible near-integrated DGP resulting from Assumption 4(ii) is motivated by the high, yet uncertain persistence of many predictor series; Assumption 4(i) additionally allows for the possibility of stationary predictors. Since persistence need not always be constant in practice, in particular when close to the unit root, we allow for time variation in persistence in the near integrated case.

Considering (1)–(2) with $(u_t, v_t)'$ serially independent Gaussian distributed innovations ($a_1 = \cdots = a_{p-1} = 0$ in Assumption 3) with mean zero and covariance matrix $\Sigma := \begin{pmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & \sigma_v^2 \end{pmatrix}$, and ρ in (2) fixed and $|\rho| < 1$, Stambaugh (1986, 1999) shows that, when $\sigma_{uv} \neq 0$, the exact OLS bias of $\hat{\beta}$ computed from (1) is $E(\hat{\beta} - \beta) = \gamma E(\hat{\rho} - \rho)$, with $\hat{\rho}$ denoting the OLS estimate of ρ and $\gamma := \sigma_{uv}/\sigma_v^2$ is the slope coefficient in a regression of u_t on v_t . Since $\hat{\rho}$ is known to be downward biased in small-samples, and $(u_t, v_t)'$ are typically strongly negatively contemporaneously correlated, the autoregressive OLS bias feeds into the small-sample distribution of $\hat{\beta}$ causing over-rejections of the null hypothesis of no predictability, $H_0 : \beta = 0$. Nelson and Kim (1993) argue that in such a context the OLS estimators and corresponding standard errors are substantially biased and lead to unreliable inference.

To correct for the OLS bias of $\hat{\beta}$, Amihud and Hurvich (2004) [AH] propose, in the context of no autocorrelation in the innovations ($p = 1$ in our setup), a simple statistical device that builds upon the OLS estimates obtained from a

¹ This is especially the case when dealing with (near-) integrated regressors; see, e.g., Cavaliere (2004) and Cavaliere et al. (2010).

² In fact, examining the proofs we give in the Appendix, it can be seen that one may relax the independence assumption to allow for weakly dependent martingale difference sequences at the cost of additional moment restrictions; however, we do not pursue this topic here and leave it for future work.

predictive regression which is augmented with estimates of v_t , the innovations to the predictor in (2); see also Lewellen (2004). Interestingly, such an approach had been proposed much earlier in the cointegration testing context by Phillips (1991), albeit in a case where the exact degree of persistence is known. In specific, regression (1) can be written under Assumption 2 as

$$y_t = \alpha + \beta x_{t-1} + \gamma v_t + \varepsilon_t \quad (3)$$

with β still capturing the extent of *ex ante* predictability in the system. However, regression (3) is infeasible since v_t is not observable. Nevertheless, it is worth focusing on (3) given the interesting properties of the resulting estimates and test statistics: this type of augmentation implies that (i) the infeasible estimator of $(\beta, \gamma)'$ in (3) is exactly unbiased; and (ii) the null distribution of the t -statistic on $\hat{\beta}$ converges asymptotically to a standard normal distribution irrespectively of the persistence of x_t and the degree of contemporaneous correlation of $(u_t, v_t)'$.

Given the nice properties just described it may be tempting to use some proxy of v_t to make this regression feasible, however the asymptotic properties of the infeasible test computed from (3) do not automatically extend to the feasible counterpart resulting from the direct use of OLS residuals from (2), say $\hat{v}_t$; see Rodrigues and Rubia (2011), Cai and Wang (2014) and Demetrescu (2014a), for further details.

The distinctive feature of the AH procedure is that it uses a *bias-adjusted* estimate of v_t to reduce the bias of $\hat{\beta}$. Thus, the resulting feasible regression is,

$$y_t^\mu = \beta x_{t-1}^\mu + \gamma \hat{v}_t^* + \varepsilon_t^*, \quad (4)$$

where ε_t^* denote the errors of the feasible regression. For the case where the putative predictor has no short run dynamics, $\hat{v}_t^* := x_t - \hat{m} - \hat{\rho}^* x_{t-1}$, with $\hat{m}$ denoting an estimated intercept in the autoregression and $\hat{\rho}^*$ a finite-sample bias-corrected OLS estimate of ρ in (2). To be specific, AH suggest the computation of this estimate as $\hat{\rho}^* = \hat{\rho} + (1 + 3\hat{\rho})/T + 3(1 + 3\hat{\rho})/T^2$, which is obtained by an iterative correction. This reduces noncentrality of the limiting distribution of the OLS t -statistic, but the distribution remains nonstandard in the near-integrated case (Assumption 4ii). The procedure also requires a correction of the standard errors which is not easily generalized to higher-order dynamics (the kind that we impose in Assumption 3); see Amihud et al. (2009, 2010).

2.2. The bias-reduced IVX approach

We therefore take a different route. The procedure we propose builds on the IVX framework of Kostakis et al. (2015). IVX centers on the construction of instrumental variables from the potential predictors, which ensures instrument relevance while controlling for persistence. In particular, for the implementation of the original IVX approach in the context of (1) and (2), Kostakis et al. use $z_t := \sum_{j=0}^{t-2} \varrho^j \Delta x_{t-j} = (1 - \varrho L)_+^{-1} \Delta x_t$ as instrument for x_t , where L denotes the conventional lag operator, and $\varrho := 1 - aT^{-\eta}$, with $0 < \eta \leq 1$, and $a \geq 0$ and fixed, and is chosen so that z_t is by construction only mildly integrated when the predictor x_t is (nearly) integrated.

The original IVX estimator of β ($\hat{\beta}^{ivx}$), computed from (1) using z_t as instrument, has a slower convergence rate than the conventional OLS estimator under near integration; concretely, the IVX convergence rate is $T^{1/2+\eta/2}$; cf. Kostakis et al. (2015). At the same time, the IVX estimator is mixed Gaussian in the limit irrespectively of the degree of endogeneity implied by γ , leading to standard inference in t and Wald tests. Moreover, under low persistence (Assumption 4i), it is asymptotically equivalent to the OLS procedure – which makes IVX easy to use in practice.

Letting $\ddot{\cdot}$ denote demeaned quantities, e.g. $\ddot{y}_t = y_t - \bar{y} = y_t - \frac{1}{T-1} \sum_{s=2}^T y_s$, the IV estimator is,

$$\hat{\beta}^{ivx} := \frac{\sum_{t=2}^T z_{t-1} \ddot{y}_t}{\sum_{t=2}^T z_{t-1} \ddot{x}_{t-1}}, \quad (5)$$

and its standard error $se(\hat{\beta}^{ivx}) := \left(\sum_{t=2}^T z_{t-1} \ddot{x}_{t-1} \right)^{-1} \hat{\sigma}_u \left(\sum_{t=2}^T z_{t-1}^2 \right)^{1/2}$. To improve finite-sample behavior, Kostakis et al. (2015) suggest the use of OLS residuals $\hat{u}_t$ from (1) (whose consistency properties do not depend on the persistence properties of the instrument z_t) for the computation of $\hat{\sigma}_u^2$, do not demean the instrument z_t (which is invariant to μ_x by construction), and correct the standard errors by subtracting from $\hat{\sigma}_u^2 \left(\sum_{t=2}^T z_{t-1}^2 \right)$ a term involving the long-run covariance of u_t and v_t .

For the implementation of the bias-reduced IVX approach in the setup given by (1) and (2), and Assumptions 1–4, the starting point is how to deal with short-run dynamics in the increments of x_t , since this has implications as to which residuals to use for augmentation of the IVX testing procedure. We let the innovations v_t correlate with u_t contemporaneously. Note that, because of the serial correlation of v_t , unrestricted correlation between u_t and v_t would not have excluded cases where v_{t-1} might have predicted u_t – while u_t is an unpredictable error term. The augmentation approach relies on the decomposition of the shocks to the predictive regression as the sum of two orthogonal components; should v_t be one of them, this induces serial correlation in u_t , which is not a plausible feature of the null hypothesis of no predictability.

As we shall see, the key feature of IVX estimation in this setup is that, since the IVX approach uses an instrument that is *less persistent* than the original predictor, it turns out that the residual augmentation approach is essentially insensitive to

whether the residuals used for augmentation are bias-corrected or not, at least in the limit; see Section 3 for details. This is important, since it simplifies the analysis considerably and allows for easy generalizations to higher order dynamics in the predictor as will be shown below.

Remark 2.1. It may be surprising that, although using OLS residuals does not work for the OLS estimation of the predictive regression, simple augmentation will work for IVX. The reason is that, the estimation noise $(\hat{v}_t - v_t)$ does not affect the IVX estimator in the limit, given the lower convergence rate of IVX compared to the OLS estimator (which would have been asymptotically affected). Moreover, the local power is the same as if the true v_t were used in the augmented regression: this reduces regression noise whenever $\gamma \neq 0$, leading by construction to a larger drift term in the distribution under the local alternative $\beta := bT^{-(1/2+\eta/2)}$; see the discussion following Theorem 3.3 for details. This may not increase the convergence rate, but considering the typically high correlation of the innovations u_t and v_t (induced by $\gamma \neq 0$), power gains in finite samples are to be expected. See also Im and Schmidt (2008) for positive power effects of residual augmentation. We examine the finite-sample power in the Monte Carlo analysis in Section 4. $\square$

The implementation of the residual-augmented IVX approach in the general framework described by Assumption 1–4 consists of the IV estimation of the feasible regression,

$$\ddot{y}_t = \beta \ddot{x}_{t-1} + \gamma \hat{v}_t + \varepsilon_t^* \quad (6)$$

where all directly observed variables are demeaned and the residuals $\hat{v}_t$ are computed from an autoregressive model of order p for the predictor x_t . E.g. if demeaning x_t prior to fitting the autoregression we would have,

$$\hat{v}_t = \ddot{x}_t - \sum_{j=1}^p \hat{\varphi}_j \ddot{x}_{t-j} = \ddot{v}_t - \sum_{j=1}^p (\hat{\varphi}_j - \varphi_j) \ddot{x}_{t-j}, \quad t = p+1, \dots, T,$$

with $\hat{\varphi}_j$, $j = 1, \dots, p$, the OLS autoregressive coefficient estimates and φ_j their probability limits. One may use some information criteria in levels to determine the autoregressive order p (in Sections 4 and 5 we use Akaike's information criteria [AIC]); note that conducting model selection in levels copes with both the stationary and the integrated cases.

Remark 2.2. The autoregression of x_t in (2) includes an intercept (i.e., is equivalently computed with demeaned variables) to account for a mean component, μ_x . The time series literature offers alternative ways of demeaning, say recursive mean adjustment as in So and Shin (1999) or, particularly appealing in settings with persistent variables is the subtraction of the initial condition (which is equivalent to local-to-unity GLS demeaning under persistence, see e.g. Elliott et al., 1996). We shall also consider these alternatives in the following sections. However, alternative demeaning schemes for y_t (and correspondingly x_{t-1} in (6)) are not such a good idea; e.g. forward/recursive demeaning (as suggested by Westerlund et al., 2017, in a panel context) eliminates the signal and leads to significant power losses; see Demetrescu and Hosseinkouchack (2020). $\square$

For simplicity of notation write (6) as

$$\ddot{y}_t = X_{t-1} \Gamma' + \varepsilon_t^* \quad (7)$$

where $X_{t-1} := (\ddot{x}_{t-1} \ \hat{v}_t)$ and $\Gamma := (\beta; \gamma)$. Moreover, define $Z_{t-1} := (z_{t-1} \ \hat{v}_t)$, where z_{t-1} is an instrument for x_{t-1} constructed as suggested by Kostakis et al. (2015) and described above. Thus, given the just-identified nature of this model the IV estimator of Γ is,

$$\hat{\Gamma} := \left(\sum_{t=2}^T Z_{t-1}' X_{t-1} \right)^{-1} \sum_{t=2}^T Z_{t-1}' \ddot{y}_t. \quad (8)$$

Since x_{t-1} and $\hat{v}_t$ are orthogonal by construction, one may compute the residual-augmented estimator and test statistic in two steps³:

1. OLS-regress $\ddot{y}_t$ on $\hat{v}_t$ and compute residuals, $\tilde{y}_t = \ddot{y}_t - \hat{\gamma} \hat{v}_t$, where $\hat{\gamma}$ is the OLS slope coefficient, $\hat{\gamma} := \left(\sum_{t=2}^T \hat{v}_t^2 \right)^{-1} \sum_{t=2}^T \hat{v}_t \ddot{y}_t$.
2. IVX-regress $\tilde{y}_t$ on $\ddot{x}_{t-1}$, leading to $\tilde{\beta}^{ivx} := \left(\sum_{t=2}^T z_{t-1} \ddot{x}_{t-1} \right)^{-1} \sum_{t=2}^T z_{t-1} \tilde{y}_t$.

Under Assumptions 1–4 the standard errors need to take into account two specific features of the data. First, time varying variances which bias the usual standard errors, even asymptotically; and second, while the estimation error $(\hat{v}_t - v_t)$ has no asymptotic effect on the limiting distribution of $\tilde{\beta}^{ivx}$ in the near-integrated context, it does so when x_t is

³ In unreported simulations (available upon request), we find this 2-step procedure to perform better in finite samples than directly estimating (6). The two versions are easily shown to be asymptotically equivalent, and we omit the details to save space.

covariance stationary. Yet treating the two cases in a different manner is inconvenient since exact knowledge about which is actually the relevant case is typically not available. Consequently, we derive heteroskedasticity-consistent standard errors for the case where ρ is bounded away from unity, and show that these are also valid in the near integrated context. In this way, we are able to use the same statistic with the same limiting distribution to cover both cases without having to decide which is which – analogously to the original IVX test of [Kostakis et al. \(2015\)](#).

To be specific, the standard error (se) is, for both variants of the residual-augmented procedure,

$$se(\tilde{\beta}^{ivx}) := \frac{\left(\sum_{t=p+1}^T z_{t-1}^2 \tilde{\varepsilon}_t^2 + \hat{\gamma}^2 \hat{Q}_T\right)^{1/2}}{\sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1}} \quad (9)$$

where the finite-sample correction $\hat{Q}_T$ is $\hat{Q}_T = \mathcal{H}_{zx} \mathcal{H}_{xx}^{-1} \mathcal{H}_{xxv} \mathcal{H}_{xx}^{-1} \mathcal{H}_{zx}$ with $\mathcal{H}_{xx} = \sum_{t=p+1}^T \ddot{x}_{t-p} (\ddot{x}_{t-p})'$, $\mathcal{H}_{zx} = \sum_{t=p+1}^T z_{t-1} (\ddot{x}_{t-p})'$, $\mathcal{H}_{xxv} = \sum_{t=p+1}^T \ddot{x}_{t-p} (\ddot{x}_{t-p})' \hat{v}_t^2$, and $\ddot{x}_{t-p} := (\ddot{x}_{t-1}, \dots, \ddot{x}_{t-p})'$. To compute the Eicker–White-type standard errors in (9) we make use of the OLS residuals computed from the residual-augmented predictive regression, $\tilde{\varepsilon}_t := \ddot{y}_t - \tilde{\beta}^{ols} \ddot{x}_{t-1} - \hat{\gamma} \hat{v}_t$, rather than IVX residuals due to the superconsistency properties of the former in the near-integrated context. The t -statistic on $\tilde{\beta}^{ivx}$ to test the null hypothesis $H_0 : \beta = 0$ in (6) is simply $\tilde{t}_{ivx} := \tilde{\beta}^{ivx} / se(\tilde{\beta}^{ivx})$.⁴

Remark 2.3. The standard errors in (9) are in fact Eicker–White standard errors that would have been appropriate under stationarity of x_t , where the effect of estimating v_t impacts the asymptotic distribution of the modified IVX t -statistic. We show in Section 3 that $\hat{Q}_T$ in (9) is dominated under near-integration so that the standard error is asymptotically equivalent to the one implied by the near-integrated framework, which is simply $\left(\sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1}\right)^{-1} \left(\sum_{t=p+1}^T z_{t-1}^2 \tilde{\varepsilon}_t^2\right)^{1/2}$. $\square$

Remark 2.4. Interestingly, the correction term $\hat{\gamma}^2 Q_T$ bears some similarities in structure with the finite-sample correction of the standard errors proposed by [Kostakis et al. \(2015\)](#) for the original IVX t and Wald statistics, with the difference that $\hat{\gamma}^2 Q_T$ is only asymptotically negligible in the near-integrated case. $\square$

Remark 2.5. A near-unit root in x_t would allow for the use of residuals without the need for the finite sample correction (see [Remark 2.4](#)), but the statistics fare better in finite samples if the correction is included (essentially because, in finite samples, any $|\rho| < 1$ is “caught between” stationarity and integration). $\square$

2.3. Extensions to multiple predictors

The discussion so far has side-stepped a couple of aspects relevant for empirical work which we address in this section. They are in fact straightforward extensions of the baseline case and we shall omit some of the technical details.

It is often the case that several predictors are considered simultaneously. Thus, the resulting multiple predictive regression framework is

$$y_t = \alpha + \beta' x_{t-1} + u_t \quad (10)$$

where x_{t-1} follows a K -dimensional vector autoregressive data generating process of order p , such as,

$$x_t - \mu_x = R(x_{t-1} - \mu_x) + v_t, \text{ and } v_t = \sum_{j=1}^{p-1} A_j v_{t-j} + v_t \quad (11)$$

which is either stable or (near) integrated as before depending on the properties of the autoregressive coefficient matrix R (v_t is taken to be a stable autoregression in either case). There is endogeneity, possibly in all regressors, expressed as a nonzero coefficient vector in the decomposition

$$u_t := \gamma' v_t + \varepsilon_t, \quad (12)$$

and the shocks v_t and ε_t are heterogeneous, serially independent obeying a multivariate version of [Assumption 2](#).

For implementation of the bias reduced IVX approach introduced in this paper in the multiple predictive regression case consider (10)–(12) and either set up the feasible multiple regression

$$\ddot{y}_t = \beta' \ddot{x}_{t-1} + \gamma' \hat{v}_t + \varepsilon_t^* \quad (13)$$

or resort to the two-step version, where the vector of residuals $\hat{v}_t$ is computed from a vector autoregression of order p of the predictors,

$$\hat{v}_t := \ddot{x}_t - \sum_{j=1}^p \hat{\Phi}_j \ddot{x}_{t-j}, \quad t = p+1, \dots, T,$$

⁴ Note that one may resort to alternative HC variance estimators, e.g., with correction for degrees of freedom (HC1). The HC1 version is obtained here by multiplying the estimated variance by $\frac{T}{T-p-3}$.

with $\hat{\Phi}_j, j = 1, \dots, p$, the matrix of OLS coefficient estimates in levels. Note that the use of AIC (or some other information criteria) in levels, for determining the order p , is again recommended.

For inference purposes note that the estimated covariance matrix of $\tilde{\beta}^{ivx}$ in this context is given by the familiar “sandwich” formula,

$$\widehat{\text{Cov}}(\tilde{\beta}^{ivx}) := \mathbf{B}_T^{-1} \mathbf{M}_T (\mathbf{B}_T^{-1})' \quad (14)$$

where $\mathbf{B}_T := \sum_{t=2}^T \mathbf{z}_{t-1} \mathbf{z}_{t-1}'$ and

$$\begin{aligned} \mathbf{M}_T := & \sum_{t=2}^T \mathbf{z}_{t-1} \mathbf{z}_{t-1}' \tilde{\varepsilon}_t^2 + \left[\boldsymbol{\gamma}' \otimes \left(\frac{1}{T} \sum_{t=2}^T \mathbf{z}_{t-1} \mathbf{x}_{t-p,K}^{\mu'} \right) \left(\sum_{t=p+1}^T \mathbf{x}_{t-p,K}^{\mu} \mathbf{x}_{t-p,K}^{\mu'} \right)^{-1} \right] \\ & \times \left(\sum_{t=p+1}^T \mathbf{v}_t \mathbf{v}_t' \otimes \mathbf{x}_{t-p,K}^{\mu} \mathbf{x}_{t-p,K}^{\mu'} \right) \left[\boldsymbol{\gamma} \otimes \left(\sum_{t=p+1}^T \mathbf{x}_{t-p,K}^{\mu} \mathbf{x}_{t-p,K}^{\mu'} \right)^{-1} \left(\frac{1}{T} \sum_{t=2}^T \mathbf{x}_{t-p,K}^{\mu} \mathbf{z}_{t-1}' \right) \right] \end{aligned}$$

with $\mathbf{x}_{t-p,K}^{\mu}$ corresponding to the vector stacking all p lags of all K demeaned regressors, i.e., $(\mathbf{x}_{t-p,K}^{\mu})' = (\ddot{x}_{t-1,1}, \dots, \ddot{x}_{t-1,K}, \ddot{x}_{t-2,1}, \dots, \ddot{x}_{t-2,K}, \dots, \ddot{x}_{t-p,1}, \dots, \ddot{x}_{t-p,K})'$.

The limiting distribution of $\tilde{\beta}^{ivx}$ is normal in the stationary case and mixed normal in the near-integrated context; the proofs are simple multivariate extensions of the results from the single-regressor case (see the following section) and will therefore be omitted. More importantly, individual and joint significance tests have their usual standard normal and χ^2 limiting distributions irrespective of the persistence and heterogeneity of the DGP as long as the robust covariance matrix estimator in (14) is used.

3. Asymptotic results

In this section, we analyze the limiting distributional characteristics of the new augmented IVX tests. Note first that the time-varying properties of the DGP, as stated in [Assumptions 1–4](#), imply different behavior in the limit compared to the stationary case. In this case, the partial sums of $\mathbf{v}_t$ and ε_t converge weakly to time-transformed Brownian motions ([Cavaliere et al., 2010](#)), i.e.

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor sT \rfloor} \begin{pmatrix} \varepsilon_t \\ \mathbf{v}_t \end{pmatrix} \Rightarrow \begin{pmatrix} \int_0^s \sigma_\varepsilon(r) dW_\varepsilon(r) \\ \int_0^s \sigma_v(r) dW_v(r) \end{pmatrix},$$

where W_ε and W_v are independent standard Wiener processes (the “classical” case is only recovered when σ_u and σ_v are constant). Let $M(s) := \int_0^s \sigma_v(r) dW_v(r)$; then, the suitably normalized regressor can be shown to converge weakly under near integration to an Ornstein–Uhlenbeck type process driven by the diffusion $M(s)$, i.e.,

$$T^{-1/2} \mathbf{x}_{\lfloor sT \rfloor} \Rightarrow \omega \int_0^s e^{-\int_r^s c(t) dt} dM(r) =: \omega X(s) \quad (15)$$

where $\omega = \left(1 - \sum_{j=1}^{p-1} a_j \right)^{-1}$; see, e.g., [Cavaliere \(2004\)](#) for the case with constant c . Moreover,

$$T^{-1/2} \ddot{\mathbf{x}}_{\lfloor sT \rfloor} \Rightarrow \omega \left(X(s) - \int_0^1 X(r) dr \right) =: \omega \ddot{X}(s). \quad (16)$$

For weak persistence, i.e., $|\rho| < 1$ and fixed, we obtain a locally stationary $\mathbf{x}_{t-1}$, and the following results can be stated.

Theorem 3.1. Under [Assumptions 1, 2, 3](#) and [4\(i\)](#), we have, as $T \rightarrow \infty$, that

$$T^{1/2} (\tilde{\beta}^{ivx} - \beta) \xrightarrow{d} \mathcal{N}(0, \sigma_\beta^2) \quad (17)$$

where

$$\sigma_\beta^2 := \frac{\alpha_0 \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds + \gamma^2 \alpha_p' \Omega^{-1} \alpha_p \int_0^1 \sigma_v^4(s) ds}{\left[\alpha_0 \int_0^1 \sigma_v^2(s) ds \right]^2} \quad (18)$$

with $\alpha_p := (\alpha_0 \dots \alpha_{p-1})'$ and $\Omega := \{\alpha_{|i-j|}\}_{1 \leq i,j \leq p}$, where $\alpha_h := \sum b_j b_{j+h}$ with b_j the moving average coefficients of x_t , $(1 - \rho L)^{-1} (1 - a_1 L - \dots - a_{p-1} L^{p-1}) = \sum_{j \geq 0} b_j L^j$. Also, $T^{1/2} se(\tilde{\beta}^{ivx}) \xrightarrow{p} \sigma_\beta$ and, under the null hypothesis, $H_0 : \beta = 0$,

$$\tilde{t}_{ivx} \xrightarrow{d} \mathcal{N}(0, 1). \quad (19)$$

The limit behavior changes under near-integration as shown in the following theorem.

Theorem 3.2. Under Assumptions 1, 2, 3 and 4(ii), we have, as $T \rightarrow \infty$, that

$$T^{1/2+\eta/2} (\tilde{\beta}^{ivx} - \beta) \Rightarrow \mathcal{MN} \left(0, \frac{a \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds}{2\omega^2 \left(X(1) \ddot{X}(1) - \int_0^1 X(s) dM(s) + \int_0^1 c(s) (X(s))^2 ds \right)^2} \right) \quad (20)$$

and

$$se(\tilde{\beta}^{ivx}) \Rightarrow \left(\frac{a}{2\omega^2} \right)^{1/2} \frac{\left(\int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds \right)^{1/2}}{X(1) \ddot{X}(1) - \int_0^1 X(s) dM(s) + \int_0^1 c(s) (X(s))^2 ds} \quad (21)$$

where $a > 0$ and $\eta \in (0, 1)$ are fixed, ω^2 plays the role of the long-run variance and is defined in (15), $\ddot{X}(s) := X(s) - \int_0^1 X(r) dr$, $X(s) := \int_0^s e^{-\int_r^s c(t) dt} \sigma_v(r) dW_v(r)$ and, $\sigma_v^2(\cdot)$ and $\sigma_\varepsilon^2(\cdot)$ are the variance functions of v_t and ε_t , respectively. Moreover, under the null hypothesis, $H_0: \beta = 0$,

$$\tilde{t}_{ivx} \Rightarrow \mathcal{N}(0, 1). \quad (22)$$

The proof of Theorem 3.2 establishes that $Q_T = o_p(T^{1+\eta})$ so that it is dominated in (9) by $\sum_{t=p+1}^T z_{t-1}^2 \tilde{\varepsilon}_t^2$ which is of exact order $O_p(T^{1+\eta})$ (see the Appendix for details), and the effect of estimating the residuals on the standard errors is negligible in the near-integrated case. Moreover, it can be seen from the proofs that, while the limiting distribution of the residual-augmented IVX based estimator is mixed Gaussian, like the one of the original IVX statistic, the asymptotic variance is different and in fact typically smaller; see Remark 3.1.

The near-integrated case is also more interesting for evaluating the local power and for a comparison with the original IVX.⁵ The power function of the residual augmented IVX is provided next.

Theorem 3.3. Under Assumptions 1, 2, 3 and 4(ii), we have for local alternatives $\beta := bT^{-(1/2+\eta/2)}$, as $T \rightarrow \infty$ that

$$\tilde{t}_{ivx} \Rightarrow \mathcal{N} \left(b \left(\frac{2\omega^2}{a} \right)^{1/2} \frac{X(1) \ddot{X}(1) - \int_0^1 X(s) dM(s) + \int_0^1 c(s) (X(s))^2 ds}{\left(\int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds \right)^{1/2}}, 1 \right). \quad (23)$$

Remark 3.1. Using the arguments in the proofs of Theorems 3.2 and 3.3, one may immediately derive the local power of the original IVX based test under near-integration,

$$t_{ivx} \Rightarrow \mathcal{N} \left(b \left(\frac{2\omega^2}{a} \right)^{1/2} \frac{X(1) \ddot{X}(1) - \int_0^1 X(s) dM(s) + \int_0^1 c(s) (X(s))^2 ds}{\left(\int_0^1 \sigma_v^2(s) \sigma_u^2(s) ds \right)^{1/2}}, 1 \right)$$

where $\sigma_u^2 = \sigma_\varepsilon^2 + \gamma^2 \sigma_v^2$. For constant variances and no short-run dynamics, this collapses to the expression derived by Breitung and Demetrescu (2015). Whenever $\gamma \neq 0$, we have that $\sigma_u^2(s) > \sigma_\varepsilon^2(s) \forall s$, and therefore $\int_0^1 \sigma_v^2(s) \sigma_u^2(s) ds > \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds$, implying that the drift term of the residual-augmented IVX statistic is larger than the drift term of the original statistic. This leads to increased power under local alternatives, at least in the limit. $\square$

While residual augmentation leads to some gains in terms of asymptotic efficiency under persistence, the augmentation also has an effect on the finite sample bias of the IVX estimator. In fact, we may identify two mechanisms. On the one hand, Demetrescu and Hosseinkouchack (2020) show that the original IVX t-statistic exhibits finite-sample noncentrality of order $O(T^{\eta/2-1/2})$. This translates into a higher-order bias of the IVX estimator by virtue of multiplication with the (positive) standard error. Now, unfeasible augmentation with v_t removes the correlation and reduces the standard error, so the leading terms of the bias would be eliminated. On the other hand, feasible residual augmentation introduces additional bias from the estimation of the autoregression. This can be seen in the proofs in the Appendix to be of the same magnitude as the first bias component, $O(T^{\eta/2-1/2})$.

Therefore, it is not clear which effect dominates; we provide simulation evidence in the following section. A summary preview of our findings in Section 4 is as follows: If the residuals are computed in an autoregression with the usual demeaning, there is no clear effect; but if resorting to alternative adjustment procedures in the autoregression (concretely, subtracting the initial condition – akin to local GLS demeaning – or recursively demeaning (see e.g. Elliott et al., 1996, and Shin and So, 2001), then the positive effect dominates and the bias of the slope coefficient estimator is reduced, in addition to its standard error. This impacts practical applications in a twofold manner. First, reduced bias implies improved forecasts; second, we find better size control for the tests based on less biased IVX estimators.

⁵ The local power in the stationary case is easily derived using standard arguments and we omit the details.

4. Finite sample performance

4.1. Monte Carlo setup

This section compares the finite sample performance of the original IVX test, which we denote as t_{ivx} , with the residual augmented version computed with residuals from an autoregressive (AR) model for the predictor under different demeaning schemes, $\tilde{t}_{ivx}^{\mu_i}$, $i = 0, 1, 2, 3$, where $\mu_0, \mu_1, \mu_2, \mu_3$ indicate the use of residuals from an AR model with no demeaning; residuals from an OLS demeaned AR; residuals from a recursively demeaned AR; and residuals from an AR in which the variables were demeaned by subtracting the initial condition, respectively.

We generate y_t and x_t as in Eqs. (1) and (2) with $p = 2$, i.e.,

$$y_t = \alpha + \beta x_{t-1} + u_t, \quad t = 2, \dots, T \quad (24)$$

$$x_t = \rho x_{t-1} + v_t, \quad v_t = a_1 v_{t-1} + e_t \quad (25)$$

with $a_1 \in \{-0.5, 0.5\}$ and $e_t \sim iid\mathcal{N}(0, 1)$. We focus on local alternatives of the form $\beta = bT^{-1}$ for two sample sizes, $T = 200$ and $T = 500$. To study the empirical size of the tests we set $b = 0$, and for the local power evaluation $b \in \{5, 10, 15, 25\}$. The persistence of the predictor is controlled by $\rho := 1 - cT^{-1}$, with $c \in \{0, 10, 20, 30, 40, 50\}$. The correlation causing endogeneity is set to -0.95 , which is not an uncommon value in practice; see, e.g., [Campbell and Yogo \(2006\)](#)⁶; recall that smaller correlations imply smaller biases and better size control anyway.

The t_{ivx} test does not require specifying the lag length, while for $\tilde{t}_{ivx}$ we choose p via Akaike's information criteria (AIC). Both t_{ivx} and $\tilde{t}_{ivx}$ are computed by demeaning the dependent variable and the regressor, but not the instrument (see Section 2 for details). Since all tests (except $\tilde{t}_{ivx}^{\mu_0}$, included for completeness) are invariant to the intercept α , we set $\alpha = 0$ without loss of generality.

We follow [Kostakis et al. \(2015\)](#) and choose $a = -1$ and $\eta = 0.95$ ($\varrho = 1 + aT^{-\eta}$) for the construction of the instruments in both. Also, we use the corrected standard errors proposed by [Kostakis et al. \(2015, pp. 1514–1515\)](#) to compute the IVX t ratio. To maintain comparability, we apply the exact same finite-sample correction for standard errors to the standard errors of the augmented IVX test statistics (additionally to the terms involving Q_T as previously discussed). Preliminary simulations (available upon request) show that $\tilde{t}_{ivx}^{\mu_1}$ performs better in terms of size with one correction term only so we do not include Q_T in the standard errors used to compute this statistic.

We employ the proposed standard errors in (9) for the computation of $\tilde{t}_{ivx}$, while, for the classical t_{ivx} , we use the HAC estimator recommended by [Kostakis et al. \(2015, pp. 1514–1515\)](#) and required by [Theorem 3.2](#).

For testing, the rejection frequencies of right-sided and two-sided tests are computed at the nominal 5% level based on 10000 Monte Carlo replications, and all results are based on standard normal critical values.

4.2. Bias analysis

[Figs. 1–5](#) present the scaled estimation bias for the residual augmented IVX estimator computed from a predictive regression augmented with residuals from an autoregressive model of the predictor with no demeaning ([Fig. 1](#)), usual demeaning ([Fig. 2](#)), adjusted by subtracting the first observation ([Fig. 3](#)) and using recursive demeaning ([Fig. 4](#)), as well as the scaled estimation bias of the IVX approach of [Kostakis et al. \(2015\)](#), ([Fig. 5](#)). To be specific, the plots refer to $TE(\tilde{\beta}^{IVX^{\mu_i}} - \beta)$, $i = 0, 1, 2, 3$, and $TE(\hat{\beta}^{IVX} - \beta)$.

We observe in [Fig. 5](#) that the original IVX estimator is biased in finite samples, with the magnitude decreasing only slowly in T . Examining [Figs. 1–4](#), it can be seen that in general the residual augmented IVX approach produces a less biased estimate of β than the conventional IVX approach. This is particularly clear from [Figs. 1, 3 and 4](#), which provide the results for residuals computed from AR models with no mean corrected predictors, with predictors for which the initial condition has been subtracted and with recursive demeaned predictors, respectively. In [Fig. 2](#), we note that the bias of the AR estimators in the autoregression of x_t (which feeds into $\hat{\beta}^{IVX^{\mu_1}}$) is large, although decreasing fast with increasing T and c ; this is not surprising since the bias of AR estimators is known to be relatively large for demeaned persistent series. Clearly, the “quality” of the residuals used for augmentation makes a difference in finite samples, and $\tilde{t}_{ivx}^{\mu_1}$ is not recommended for testing.

4.3. Forecast risk

In this section we further evaluate the different approaches in a forecasting context where evaluation focuses on minimizing risk. To be specific, we use the one-step-ahead expected squared forecast error suggested by [Hansen \(2010\)](#) in a similar autoregressive context.

To study the forecast performance of residual augmented IVX relative to the original IVX procedure, we generate in a first experiment (Experiment 1) data from (24) and (25) with $a_1 \in \{-0.5, 0, 0.5\}$ and $e_t \sim iid\mathcal{N}(0, 1)$. With no loss of

⁶ The efficient tests of [Campbell and Yogo \(2006\)](#) (denoted as CY) and the residual augmented predictive regression based test of [Amihud et al. \(2010\)](#) (denoted as AHW) have also been analyzed and results can be found in the accompanying working paper.

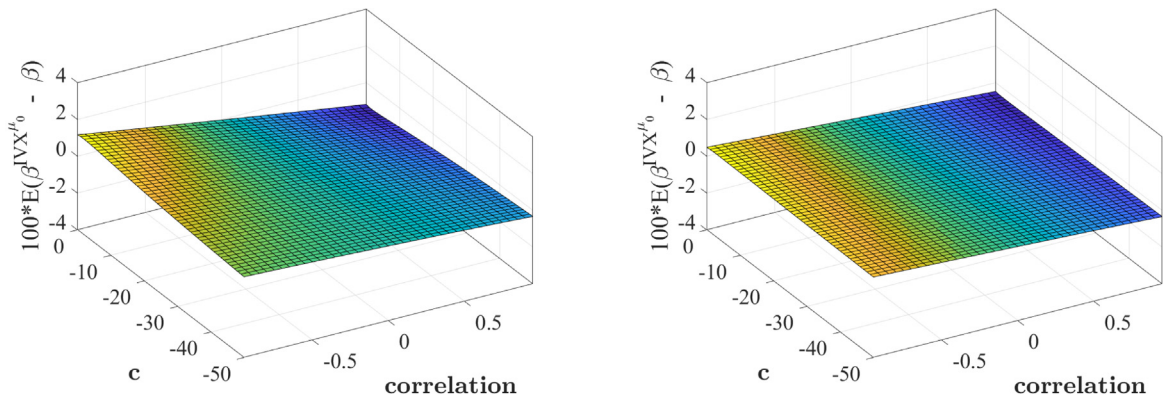


Fig. 1. Bias of residual IVX (augmented regression with residuals from an autoregressive model with no demeaning) estimator of β – $T = 200$ (left) and $T = 500$ (right).

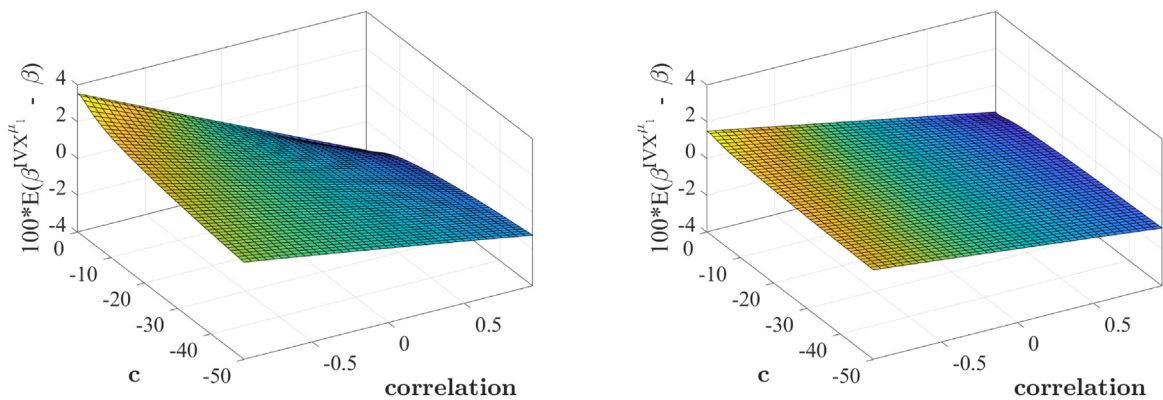


Fig. 2. Bias of residual IVX (augmented regression with residuals from an OLS demeaned autoregressive model) estimator of β – $T = 200$ (left) and $T = 500$ (right).

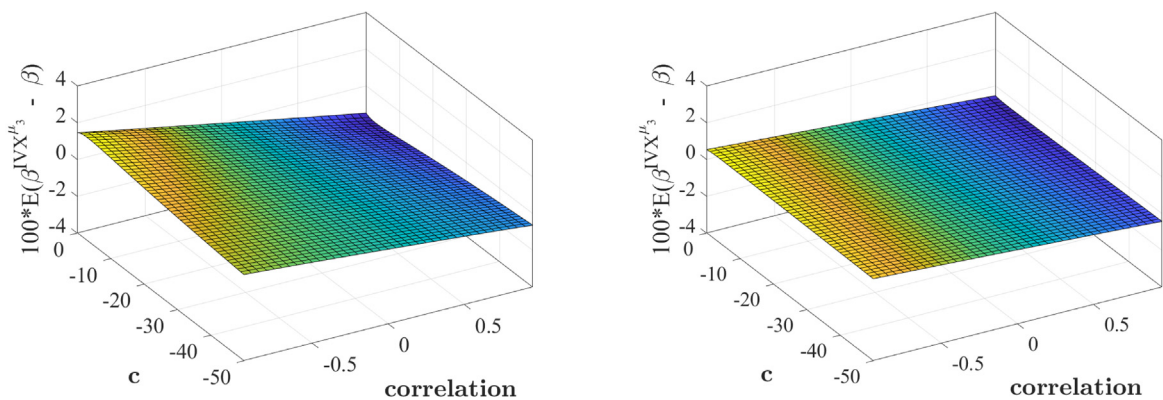


Fig. 3. Bias of residual IVX (augmented regression with residuals from an autoregressive model corrected for the first observation ($x_t - x_1$)) estimator of β – $T = 200$ (left) and $T = 500$ (right).

generality we set $\alpha = \beta = 0$ in (24) and consider $T = 200$ and $T = 500$. Regarding the persistence of the predictor we use $\rho := 1 - cT^{-1}$, with $c \in \{0, 2.5, 5, 10, 20\}$. The innovations correlation is set to -0.95 .

To evaluate the out-of-sample prediction performances of the various approaches discussed above, we estimate the predictive regression in (24) using $T - 1$ observations; for the computation of the finite sample forecast risk we take as

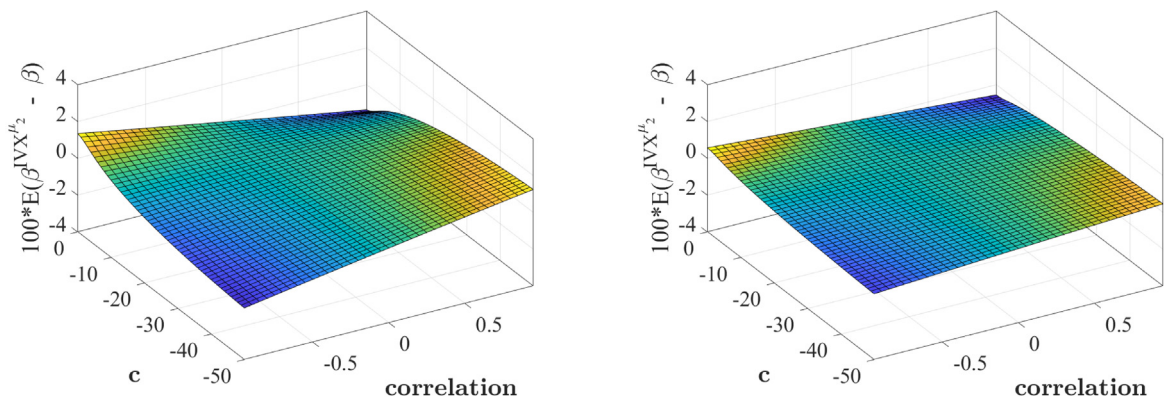


Fig. 4. Bias of residual IVX estimator (augmented regression with residuals from a recursively demeaned autoregressive model) β $-T = 200$ (left) and $T = 500$ (right).

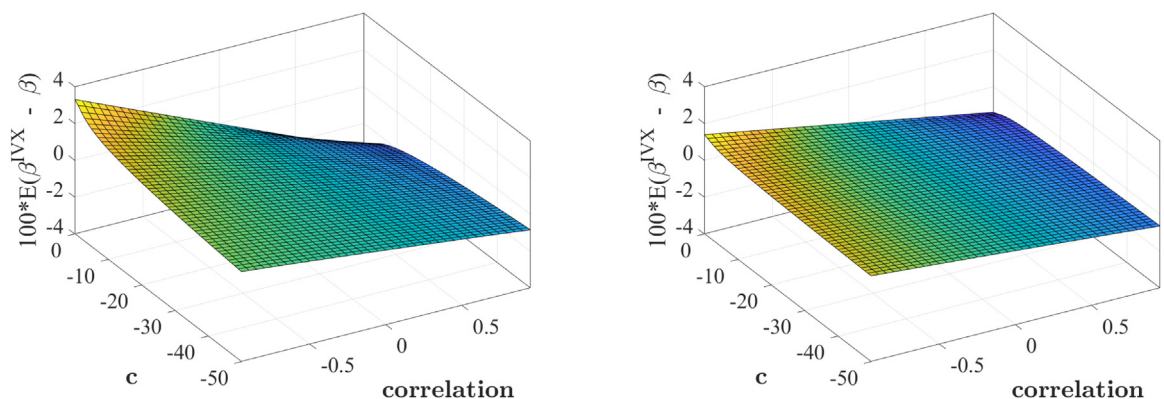


Fig. 5. Bias of IVX estimator of β $-T = 200$ (left) and $T = 500$ (right).

discussed in Hansen (2010) the average of $MSE_{\mu_k} = T(x_{T-1}(\beta - \hat{\beta}_{ivx}^{\mu_k}))^2$, $k = 0, 1, 2, 3$ and of $MSE_{ivx} = T(x_{T-1}(\beta - \hat{\beta}_{ivx}))^2$ across the 10000 Monte Carlo replications.⁷

In Table 1 we report the results for the relative forecast risk of the different approaches when benchmarked against the IVX approach. Specifically, this table reports the MSE ratios.

Table 1 shows that as c decreases the forecast risk of the residual-augmented IVX relative to IVX decreases considerably, particularly when no demeaning, recursive demeaning and subtraction of initial condition is considered (MSE_{μ_0}/MSE_{ivx} , MSE_{μ_2}/MSE_{ivx} , and MSE_{μ_3}/MSE_{ivx}) and regardless of the sample size. Overall, the smallest values (by quite a large margin, especially under persistence) are those corresponding to MSE_{μ_0}/MSE_{ivx} and MSE_{μ_3}/MSE_{ivx} . This table shows that the original IVX does roughly as well as when the residual-augmented approach with OLS demeaned residuals is used, which is by far the worst performing approach of the demeaning alternatives considered. The other approaches almost uniformly outperform the IVX and often by a very wide margin.

It is also observed that this pattern is visible regardless of whether the predictors' innovations are autocorrelated or not and regardless of whether autocorrelation is positive or negative. The gains over IVX also increase with T .

To further evaluate the forecast risk performance of the procedures, in a second experiment (Experiment 2) we generate data considering a more adverse environment allowing for the empirically relevant features described below in Section 4.5. In Table 2 we report the results for MSE_{μ_k}/MSE_{ivx} , $k = 0, 1, 2, 3$ obtained based on DGPs with time-varying volatility and the time-varying persistence patterns detailed in Section 4.5.

The first observation we can make from Table 2 is that a persistence change keeping the variance constant produces basically similar forecasting risk ratios as those observed in Table 1. Overall, MSE_{μ_1}/MSE_{ivx} suggests that the procedure using residuals obtained from an OLS demeaned regression displays larger or equal forecast risk to that of the IVX except when the break in c is towards high persistence (which is natural given that high persistence implies predictors x_{T-1}

⁷ We note that Hansen (2010) works with the asymptotics of the forecast risk, but for near-integrated processes. We choose here to work with the finite-sample counterparts for IVX and residual-augmented IVX. Asymptotic statements may be based on the results from Section 3, but we omit the details to save space.

Table 1
Forecast risk ratio (Experiment 1).

a_1	c	T = 200				T = 500			
		$\frac{MSE_{\mu_0}}{MSE_{ivx}}$	$\frac{MSE_{\mu_1}}{MSE_{ivx}}$	$\frac{MSE_{\mu_2}}{MSE_{ivx}}$	$\frac{MSE_{\mu_3}}{MSE_{ivx}}$	$\frac{MSE_{\mu_0}}{MSE_{ivx}}$	$\frac{MSE_{\mu_1}}{MSE_{ivx}}$	$\frac{MSE_{\mu_2}}{MSE_{ivx}}$	$\frac{MSE_{\mu_3}}{MSE_{ivx}}$
−0.5	20	0.871	1.016	1.048	0.887	0.808	1.000	1.000	0.808
	10	0.713	1.025	0.900	0.750	0.647	1.029	0.882	0.647
	5	0.492	0.992	0.697	0.523	0.439	0.965	0.649	0.439
	2.5	0.311	0.914	0.486	0.331	0.279	0.874	0.441	0.279
	0	0.144	0.751	0.219	0.160	0.125	0.713	0.201	0.132
0	20	0.833	1.015	1.000	0.848	0.808	1.038	1.038	0.808
	10	0.667	1.023	0.851	0.701	0.629	1.029	0.857	0.657
	5	0.442	0.980	0.653	0.476	0.417	0.950	0.633	0.417
	2.5	0.272	0.895	0.446	0.293	0.267	0.862	0.431	0.267
	0	0.125	0.743	0.201	0.141	0.117	0.710	0.190	0.124
0.5	20	0.812	1.029	0.971	0.841	0.815	1.037	1.000	0.815
	10	0.645	1.022	0.828	0.677	0.622	1.000	0.838	0.622
	5	0.430	0.975	0.639	0.462	0.403	0.952	0.613	0.419
	2.5	0.268	0.902	0.444	0.291	0.258	0.858	0.425	0.258
	0	0.124	0.737	0.206	0.141	0.117	0.707	0.190	0.124

Table 2
Forecast risk ratio (Experiment 2).

$c(s)$	$\sigma_\varepsilon^2 = \sigma_v^2$	T = 200				T = 500			
		$\frac{MSE_{\mu_0}}{MSE_{ivx}}$	$\frac{MSE_{\mu_1}}{MSE_{ivx}}$	$\frac{MSE_{\mu_2}}{MSE_{ivx}}$	$\frac{MSE_{\mu_3}}{MSE_{ivx}}$	$\frac{MSE_{\mu_0}}{MSE_{ivx}}$	$\frac{MSE_{\mu_1}}{MSE_{ivx}}$	$\frac{MSE_{\mu_2}}{MSE_{ivx}}$	$\frac{MSE_{\mu_3}}{MSE_{ivx}}$
5	1	0.481	0.993	0.674	0.504	0.448	0.966	0.655	0.466
	1 + 8I ($s > 0.3$)	0.693	0.935	0.792	0.705	0.661	0.921	0.777	0.663
	1 + 8II ($s > 0.7$)	0.867	0.992	0.978	0.867	0.849	0.986	0.959	0.851
	9 − 8I ($s > 0.3$)	0.579	1.139	0.602	0.643	0.504	1.145	0.556	0.530
	9 − 8II ($s > 0.7$)	0.467	1.127	0.609	0.522	0.412	1.118	0.588	0.437
5 + 5II ($s > 0.5$)	1	0.570	1.101	0.772	0.620	0.529	1.088	0.765	0.559
	1 + 8II ($s > 0.3$)	0.730	0.988	0.850	0.755	0.690	0.976	0.833	0.699
	1 + 8II ($s > 0.7$)	0.887	1.012	0.988	0.883	0.860	1.013	0.969	0.858
	9 − 8II ($s > 0.3$)	0.738	1.163	0.638	0.816	0.683	1.183	0.600	0.733
	9 − 8II ($s > 0.7$)	0.673	1.191	0.673	0.755	0.630	1.217	0.652	0.674
5 + 20II ($s > 0.5$)	1	0.725	1.200	0.825	0.800	0.706	1.176	0.765	0.706
	1 + 8II ($s > 0.3$)	0.809	1.051	0.906	0.851	0.765	1.056	0.898	0.786
	1 + 8II ($s > 0.7$)	0.928	1.039	0.997	0.925	0.889	1.041	0.973	0.885
	9 − 8II ($s > 0.3$)	0.877	1.178	0.644	0.945	0.828	1.172	0.586	0.862
	9 − 8II ($s > 0.7$)	0.846	1.192	0.712	0.904	0.762	1.190	0.667	0.810
20	1	0.895	1.018	1.018	0.912	0.840	1.000	1.000	0.840
	1 + 8II ($s > 0.3$)	0.918	0.991	0.957	0.921	0.911	0.996	0.955	0.907
	1 + 8II ($s > 0.7$)	0.983	1.008	1.004	0.974	0.967	1.005	0.981	0.960
	9 − 8II ($s > 0.3$)	0.957	1.061	1.052	0.983	0.920	1.060	0.940	0.920
	9 − 8II ($s > 0.7$)	0.911	1.063	0.937	0.937	0.853	1.088	0.912	0.882
25 − 5II ($s > 0.5$)	1	0.889	1.016	1.016	0.889	0.857	1.000	1.000	0.857
	1 + 8II ($s > 0.3$)	0.907	0.982	0.953	0.908	0.902	0.986	0.949	0.898
	1 + 8II ($s > 0.7$)	0.981	1.005	1.004	0.970	0.965	1.002	0.982	0.960
	9 − 8II ($s > 0.3$)	0.939	1.053	1.053	0.962	0.895	1.070	0.965	0.912
	9 − 8II ($s > 0.7$)	0.889	1.067	0.922	0.922	0.846	1.077	0.897	0.872
25 − 20II ($s > 0.5$)	1	0.877	0.951	0.992	0.836	0.870	0.944	0.981	0.833
	1 + 8II ($s > 0.3$)	0.894	0.946	0.935	0.875	0.883	0.939	0.921	0.871
	1 + 8II ($s > 0.7$)	0.994	0.998	1.021	0.991	0.992	0.997	1.008	0.994
	9 − 8II ($s > 0.3$)	0.882	0.997	1.081	0.882	0.789	0.970	1.000	0.789
	9 − 8II ($s > 0.7$)	0.694	0.953	0.795	0.701	0.646	0.929	0.772	0.638

of large magnitude, such that even smaller improvements in the quality of the residual-augmented estimate of β are affecting the forecast risk). Alternative demeaning schemes are again leading to best performing residual-augmented IVX estimators and forecasts, as seen by MSE_{μ_0}/MSE_{ivx} and MSE_{μ_3}/MSE_{ivx} typically displaying the lowest values.

The case corresponding to a large increase of the variance late in the sample ($\sigma_\varepsilon^2 = \sigma_v^2 = 1 + 8I(s > 0.7)$) shows a substantial increase in the MSE_{μ_0}/MSE_{ivx} , MSE_{μ_2}/MSE_{ivx} and MSE_{μ_3}/MSE_{ivx} ratios, but no change of the ranking.

Table 3

Empirical rejection frequencies of **right-sided tests** against local alternatives (negative short-run AR parameter ($a_1 = -0.5$) and error correlation of -0.95).

	b	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
c = 0	0	0.116	0.054	0.102	0.046	0.066	0.118	0.056	0.104	0.050	0.060
	5	0.623	0.283	0.514	0.215	0.322	0.612	0.288	0.491	0.212	0.305
	10	0.965	0.646	0.912	0.724	0.683	0.960	0.652	0.902	0.726	0.664
	15	0.997	0.842	0.981	0.916	0.861	0.995	0.854	0.973	0.921	0.860
	25	1.000	0.965	0.998	0.987	0.969	1.000	0.968	0.996	0.988	0.970
c = 10	0	0.088	0.055	0.089	0.044	0.065	0.096	0.056	0.095	0.047	0.060
	5	0.236	0.162	0.238	0.128	0.187	0.238	0.161	0.237	0.125	0.171
	10	0.515	0.411	0.527	0.317	0.460	0.506	0.395	0.509	0.304	0.418
	15	0.863	0.770	0.877	0.659	0.819	0.855	0.757	0.867	0.639	0.783
	25	1.000	0.990	1.000	0.999	0.996	1.000	0.989	1.000	0.999	0.993
c = 20	0	0.074	0.055	0.075	0.038	0.062	0.084	0.056	0.083	0.042	0.061
	5	0.169	0.130	0.175	0.095	0.147	0.172	0.128	0.174	0.091	0.135
	10	0.347	0.285	0.354	0.213	0.313	0.333	0.267	0.335	0.197	0.280
	15	0.592	0.523	0.603	0.407	0.561	0.564	0.489	0.572	0.372	0.507
	25	0.966	0.943	0.971	0.880	0.959	0.960	0.931	0.962	0.860	0.944
c = 30	0	0.066	0.053	0.069	0.035	0.058	0.079	0.057	0.078	0.040	0.060
	5	0.145	0.117	0.148	0.083	0.129	0.148	0.118	0.149	0.082	0.123
	10	0.280	0.237	0.286	0.172	0.260	0.266	0.215	0.265	0.154	0.225
	15	0.471	0.418	0.480	0.322	0.449	0.435	0.380	0.438	0.281	0.394
	25	0.860	0.825	0.865	0.727	0.847	0.827	0.781	0.832	0.661	0.798
c = 40	0	0.064	0.050	0.065	0.033	0.055	0.074	0.057	0.074	0.039	0.061
	5	0.133	0.112	0.137	0.078	0.121	0.134	0.111	0.135	0.074	0.115
	10	0.246	0.213	0.251	0.153	0.229	0.230	0.189	0.230	0.135	0.197
	15	0.409	0.365	0.416	0.282	0.388	0.362	0.316	0.366	0.233	0.329
	25	0.771	0.734	0.779	0.631	0.756	0.706	0.662	0.712	0.540	0.676
c = 50	0	0.061	0.050	0.063	0.033	0.055	0.071	0.057	0.071	0.038	0.060
	5	0.119	0.106	0.125	0.075	0.113	0.125	0.105	0.125	0.071	0.110
	10	0.226	0.199	0.230	0.145	0.213	0.206	0.176	0.207	0.125	0.182
	15	0.370	0.337	0.376	0.257	0.354	0.318	0.280	0.322	0.205	0.290
	25	0.705	0.676	0.713	0.571	0.692	0.620	0.579	0.627	0.465	0.592

Notes: t_{ivx} is the IVX test computed following [Kostakis et al. \(2015\)](#) and $\tilde{t}_{ivx}^{\mu_i}$, $i = 0, 1, 2, 3$ the residual-augmented IVX test procedure introduced in this paper, all with maximal lag length $p = \lfloor 4(T/100)^{0.25} \rfloor$ and where $\tilde{t}_{ivx}^{\mu_0}$ indicates that residuals from an AR model with no demeaning are used; $\tilde{t}_{ivx}^{\mu_1}$ residuals from an OLS demeaned AR are used; $\tilde{t}_{ivx}^{\mu_2}$ residuals from a recursively demeaned AR are used; and $\tilde{t}_{ivx}^{\mu_3}$ residuals from an AR based on variables which had the initial condition removed, are used. The DGP is as in (24) and (25) with $\rho = 1 - cT^{-1}$ and $\beta = bT^{-1}$.

4.4. Empirical rejection frequencies

To evaluate the in-sample rejection frequencies of the different procedures, we focus on one-sided testing (right-sided, more specifically), since the sign of the slope coefficients is often known and dictated by economic theory, but for completeness we also provide results for two-sided tests. [Table 3](#) presents the results for the right-sided tests obtained when v_t follows an AR(1) with $a_1 = -0.5$ (negative autocorrelation) in (25).⁸ This is especially the case when c is small (see [Demetrescu and Hosseinkouchack, 2020](#), for a discussion of the IVX test statistic in one-sided vs. two-sided test setups). From [Table 3](#) we observe that for $T = 200$ ($T = 500$), when $b = 0$, t_{ivx} and $\tilde{t}_{ivx}^{\mu_1}$ are oversized. Note that the empirical rejection frequencies for the t_{ivx} test when $c = (0, 10, 20)$ for $T = 200$ and $T = 500$ are (11.6%, 8.8%, 7.4%) and (11.8%, 9.6%, 8.4%), respectively, whereas for $\tilde{t}_{ivx}^{\mu_1}$ the corresponding rejection frequencies are (10.2%, 8.9%, 7.5%) and (10.4%, 9.5%, 8.3%), respectively. At the same time, we also observe that $\tilde{t}_{ivx}^{\mu_0}$, $\tilde{t}_{ivx}^{\mu_2}$ and $\tilde{t}_{ivx}^{\mu_3}$ display quite acceptable rejection frequencies (for these statistics the empirical size is between [3.3%, 6.6%] for $T = 200$ and [3.8%, 6.1%] for $T = 500$). In this experiment $\tilde{t}_{ivx}^{\mu_1}$ presents the largest size distortions of the residual augmented tests as a consequence of the OLS demeaning of the AR model considered in computing the residuals. Regarding power we observe that out of the residual augmented tests with adequate empirical size control ($\tilde{t}_{ivx}^{\mu_0}$, $\tilde{t}_{ivx}^{\mu_2}$ and $\tilde{t}_{ivx}^{\mu_3}$), $\tilde{t}_{ivx}^{\mu_3}$ displays the best performance and it also compares relatively well to the t_{ivx} , although the latter may display oversized behavior.

⁸ In the [Appendix](#) we also provide results for positive short-run dynamics, i.e., when $a_1 = 0.5$ (see [Tables B.1–B.2](#)). We observe in general some size distortions for all tests, with t_{ivx} displaying the most severe distortions when compared to the other procedures, and $\tilde{t}_{ivx}^{\mu_i}$, $i = 0, 2, 3$ display the smallest distortions.

Table 4

Empirical rejection frequencies of **two-sided tests** against local alternatives (negative short-run AR parameter ($a_1 = -0.5$) and error correlation of -0.95).

b		t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
c = 0	0	0.056	0.029	0.051	0.026	0.034	0.056	0.029	0.054	0.028	0.032
	5	0.359	0.196	0.313	0.119	0.223	0.338	0.195	0.297	0.119	0.209
	10	0.892	0.530	0.822	0.572	0.572	0.883	0.537	0.802	0.571	0.552
	15	0.984	0.759	0.955	0.849	0.787	0.978	0.778	0.940	0.851	0.787
	25	0.999	0.933	0.995	0.973	0.941	0.999	0.943	0.990	0.974	0.945
c = 10	0	0.056	0.036	0.056	0.067	0.039	0.057	0.034	0.055	0.065	0.036
	5	0.137	0.093	0.140	0.072	0.108	0.138	0.089	0.137	0.073	0.097
	10	0.357	0.264	0.358	0.199	0.298	0.344	0.251	0.342	0.187	0.267
	15	0.723	0.613	0.739	0.482	0.669	0.708	0.594	0.713	0.458	0.621
	25	1.000	0.976	1.000	0.992	0.988	1.000	0.972	1.000	0.991	0.979
c = 20	0	0.054	0.044	0.053	0.072	0.045	0.054	0.041	0.054	0.071	0.041
	5	0.100	0.077	0.103	0.064	0.085	0.104	0.072	0.104	0.063	0.076
	10	0.229	0.178	0.232	0.129	0.200	0.215	0.164	0.215	0.116	0.173
	15	0.434	0.367	0.446	0.278	0.404	0.406	0.336	0.410	0.246	0.353
	25	0.913	0.869	0.920	0.776	0.896	0.896	0.843	0.901	0.729	0.859
c = 30	0	0.053	0.048	0.051	0.074	0.049	0.057	0.045	0.055	0.069	0.046
	5	0.087	0.070	0.090	0.062	0.075	0.091	0.066	0.089	0.064	0.071
	10	0.178	0.146	0.183	0.107	0.161	0.167	0.133	0.168	0.095	0.139
	15	0.334	0.287	0.342	0.210	0.311	0.299	0.246	0.301	0.179	0.258
	25	0.756	0.704	0.766	0.592	0.734	0.696	0.639	0.702	0.514	0.657
c = 40	0	0.052	0.051	0.051	0.073	0.050	0.055	0.046	0.054	0.068	0.047
	5	0.081	0.065	0.082	0.063	0.072	0.082	0.066	0.082	0.062	0.069
	10	0.152	0.132	0.157	0.099	0.142	0.142	0.120	0.143	0.086	0.124
	15	0.281	0.247	0.287	0.181	0.267	0.244	0.204	0.246	0.145	0.211
	25	0.643	0.606	0.654	0.494	0.631	0.563	0.515	0.570	0.399	0.529
c = 50	0	0.051	0.050	0.051	0.072	0.050	0.053	0.047	0.052	0.067	0.048
	5	0.076	0.064	0.077	0.063	0.067	0.077	0.063	0.078	0.061	0.066
	10	0.140	0.122	0.145	0.093	0.130	0.129	0.109	0.128	0.080	0.113
	15	0.255	0.224	0.260	0.167	0.240	0.212	0.182	0.212	0.130	0.188
	25	0.578	0.541	0.587	0.440	0.559	0.481	0.439	0.489	0.336	0.450

Notes: t_{ivx} is the IVX test computed following [Kostakis et al. \(2015\)](#) and $\tilde{t}_{ivx}^{\mu_i}$, $i = 0, 1, 2, 3$ the residual-augmented IVX test procedure introduced in this paper, all with maximal lag length $p = \lfloor 4(T/100)^{0.25} \rfloor$ and where $\tilde{t}_{ivx}^{\mu_0}$ indicates that residuals from an AR model with no demeaning are used; $\tilde{t}_{ivx}^{\mu_1}$ residuals from an OLS demeaned AR are used; $\tilde{t}_{ivx}^{\mu_2}$ residuals from a recursively demeaned AR are used; and $\tilde{t}_{ivx}^{\mu_3}$ residuals from an AR based on variables which had the initial condition removed, are used. The DGP is as in (24) and (25) with $\rho = 1 - cT^{-1}$ and $\beta = bT^{-1}$.

In what concerns the two-sided versions of the statistics we observe from [Table 4](#) that all tests with the exception of $\tilde{t}_{ivx}^{\mu_2}$ display acceptable size control (although $\tilde{t}_{ivx}^{\mu_0}$ and $\tilde{t}_{ivx}^{\mu_3}$ display more conservative behavior when the predictor is more persistent). We further observe that in the homogeneous framework considered in the simulations which led to the results in [Table 4](#), and which only considers right-tailed alternatives, that the t_{ivx} displays in general the best power performance, though not by a large margin compared to $\tilde{t}_{ivx}^{\mu_1}$.⁹

4.5. Robustness against empirical features of the data

To evaluate the performance of the procedures under other empirically relevant features we report results for the empirical size under DGPs with time-varying volatility and time-varying persistence. To be specific, we consider five common variance patterns:

1. constant, $\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 1$;
2. an early upward break, $\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 1 + 8\mathbb{I}(s > 0.3)$;
3. a late upward break, $\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 1 + 8\mathbb{I}(s > 0.7)$;
4. an early downward break, $\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 9 - 8\mathbb{I}(s > 0.3)$;
5. a late downward break, $\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 9 - 8\mathbb{I}(s > 0.7)$,

⁹ Results available from the authors show that in a context where left-sided alternatives are considered both the left-sided and two-sided tests computed from residual-augmented predictive regressions display for all cases superior power performance than the t_{ivx} .

Table 5

Empirical rejection frequencies of **right-sided tests** under breaks in variance and persistence, negative short-run AR parameter ($a_1 = -0.5$) and error correlation of -0.95 .

$c(s)$	$\sigma_\varepsilon^2 = \sigma_v^2$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
5	1	0.109	0.058	0.105	0.053	0.069	0.114	0.062	0.111	0.055	0.066
	1 + 8 $\mathbb{I}(s > 0.3)$	0.140	0.063	0.121	0.061	0.073	0.141	0.064	0.122	0.062	0.068
	1 + 8 $\mathbb{I}(s > 0.7)$	0.197	0.075	0.180	0.072	0.084	0.196	0.067	0.177	0.067	0.071
	9 – 8 $\mathbb{I}(s > 0.3)$	0.178	0.069	0.199	0.051	0.078	0.185	0.067	0.205	0.053	0.072
	9 – 8 $\mathbb{I}(s > 0.7)$	0.138	0.063	0.144	0.055	0.073	0.133	0.062	0.135	0.052	0.065
5 + 5 $\mathbb{I}(s > 0.5)$	1	0.100	0.057	0.112	0.053	0.066	0.105	0.061	0.112	0.055	0.065
	1 + 8 $\mathbb{I}(s > 0.3)$	0.129	0.064	0.121	0.063	0.072	0.130	0.061	0.121	0.062	0.066
	1 + 8 $\mathbb{I}(s > 0.7)$	0.177	0.071	0.170	0.071	0.079	0.179	0.063	0.170	0.066	0.066
	9 – 8 $\mathbb{I}(s > 0.3)$	0.179	0.075	0.205	0.051	0.086	0.184	0.073	0.213	0.055	0.076
	9 – 8 $\mathbb{I}(s > 0.7)$	0.126	0.066	0.149	0.055	0.075	0.126	0.069	0.146	0.055	0.072
5 + 20 $\mathbb{I}(s > 0.5)$	1	0.090	0.057	0.109	0.051	0.067	0.094	0.061	0.112	0.049	0.065
	1 + 8 $\mathbb{I}(s > 0.3)$	0.110	0.060	0.114	0.060	0.069	0.116	0.059	0.118	0.063	0.062
	1 + 8 $\mathbb{I}(s > 0.7)$	0.144	0.064	0.149	0.060	0.071	0.147	0.056	0.151	0.060	0.060
	9 – 8 $\mathbb{I}(s > 0.3)$	0.180	0.082	0.208	0.049	0.091	0.188	0.081	0.218	0.051	0.086
	9 – 8 $\mathbb{I}(s > 0.7)$	0.120	0.069	0.142	0.049	0.076	0.123	0.072	0.147	0.049	0.075
20	1	0.076	0.058	0.078	0.039	0.064	0.078	0.056	0.079	0.039	0.060
	1 + 8 $\mathbb{I}(s > 0.3)$	0.103	0.058	0.099	0.054	0.067	0.108	0.059	0.101	0.056	0.063
	1 + 8 $\mathbb{I}(s > 0.7)$	0.165	0.070	0.159	0.068	0.076	0.172	0.063	0.163	0.058	0.066
	9 – 8 $\mathbb{I}(s > 0.3)$	0.144	0.067	0.154	0.038	0.072	0.162	0.073	0.172	0.038	0.076
	9 – 8 $\mathbb{I}(s > 0.7)$	0.103	0.063	0.111	0.044	0.069	0.104	0.064	0.110	0.041	0.067
25 – 5 $\mathbb{I}(s > 0.5)$	1	0.079	0.056	0.078	0.042	0.065	0.080	0.056	0.079	0.040	0.060
	1 + 8 $\mathbb{I}(s > 0.3)$	0.106	0.060	0.100	0.056	0.068	0.112	0.060	0.106	0.058	0.063
	1 + 8 $\mathbb{I}(s > 0.7)$	0.172	0.072	0.163	0.068	0.078	0.180	0.066	0.168	0.061	0.069
	9 – 8 $\mathbb{I}(s > 0.3)$	0.143	0.065	0.151	0.038	0.070	0.160	0.071	0.168	0.037	0.074
	9 – 8 $\mathbb{I}(s > 0.7)$	0.104	0.062	0.111	0.045	0.068	0.104	0.064	0.108	0.041	0.067
25 – 20 $\mathbb{I}(s > 0.5)$	1	0.091	0.051	0.074	0.043	0.061	0.102	0.055	0.081	0.045	0.060
	1 + 8 $\mathbb{I}(s > 0.3)$	0.132	0.060	0.109	0.057	0.070	0.138	0.063	0.114	0.059	0.066
	1 + 8 $\mathbb{I}(s > 0.7)$	0.214	0.081	0.188	0.075	0.089	0.214	0.076	0.186	0.069	0.079
	9 – 8 $\mathbb{I}(s > 0.3)$	0.134	0.054	0.121	0.034	0.061	0.139	0.057	0.124	0.035	0.061
	9 – 8 $\mathbb{I}(s > 0.7)$	0.115	0.054	0.099	0.045	0.062	0.114	0.053	0.095	0.041	0.057

Notes: See notes under Table 1. The DGP is as in (1) and (2) with $\rho = 1 - c_T T^{-1}$ and $\beta = 0$ and exhibits time-varying variances. For further details see the text.

where $\mathbb{I}(\cdot)$ is an indicator function; and to allow for time-varying persistence, six patterns for the localization parameter c are considered:

1. constant close to integration, $c(s) = 5$;
2. small break towards stationarity, $c(s) = 5 + 5\mathbb{I}(s > 0.5)$;
3. large break towards stationarity, $c(s) = 5 + 20\mathbb{I}(s > 0.5)$;
4. constant close to stationarity, $c(s) = 25$;
5. small break towards integration, $c(s) = 25 - 5\mathbb{I}(s > 0.5)$;
6. large break towards integration, $c(s) = 25 - 20\mathbb{I}(s > 0.5)$.

Table 5 provides the results for the right-sided tests computed from data with the features described above. We observe that the oversizing of t_{ivx} noted in the homogeneous DGP context in the previous section, deteriorates even further in the presence of variance breaks. From Table 5 we note that for $T = 200$ ($T = 500$) the rejection frequencies of t_{ivx} are between [7.9%, 21.4%] ([7.8%, 21.4%]), with the worse situation generally occurring when a late upward break in variance is observed ($\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 1 + 8\mathbb{I}(s > 0.7)$). This is explained by the findings of Demetrescu and Hosseinkouchack (2020) who show higher-order noncentrality of the IVX t statistic (and thus size distortions) to depend on the variance pattern.

On the other hand, the size control of $\tilde{t}_{ivx}^{\mu_0}$, $\tilde{t}_{ivx}^{\mu_2}$ and $\tilde{t}_{ivx}^{\mu_3}$ is overall quite good. We observe that in general the rejection frequencies of these tests are between [4.3%, 8.9%] for $T = 200$, and [5.3%, 8.6%] for $T = 500$, with the largest rejection frequencies being observed for $\tilde{t}_{ivx}^{\mu_3}$. $\tilde{t}_{ivx}^{\mu_1}$ is again an exception when compared to the other residual augmented approaches as it displays similar oversizing patterns as t_{ivx} .

Regarding the two-sided tests, Table 6 shows that results, overall, improve somewhat. Note that for $T = 200$ ($T = 500$) the rejection frequencies of the two-sided t_{ivx} are between [5.4%, 17.2%] ([5.5%, 17.1%]), with the worse situation again occurring when a late upward break in variance is observed. For $\tilde{t}_{ivx}^{\mu_1}$ the rejection frequencies are very similar to those of the two-sided t_{ivx} test, i.e., [4.6%, 16.0%] for $T = 200$ and [4.9%, 15.8%] for $T = 500$. For the other two-sided residual

Table 6

Empirical rejection frequencies of **two-sided tests** under breaks in variance and persistence, negative short-run AR parameter ($a_1 = -0.5$) and error correlation of -0.95 .

$c(s)$	$\sigma_\varepsilon^2 = \sigma_v^2$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
5	1	0.061	0.030	0.057	0.046	0.035	0.066	0.033	0.062	0.054	0.036
	1 + 8I ($s > 0.3$)	0.087	0.034	0.075	0.039	0.039	0.089	0.035	0.074	0.037	0.037
	1 + 8I ($s > 0.7$)	0.154	0.047	0.143	0.051	0.054	0.150	0.040	0.135	0.041	0.043
	9 – 8I ($s > 0.3$)	0.127	0.041	0.132	0.116	0.044	0.127	0.037	0.134	0.126	0.038
	9 – 8I ($s > 0.7$)	0.080	0.033	0.082	0.056	0.038	0.080	0.033	0.080	0.063	0.035
5 + 5I ($s > 0.5$)	1	0.061	0.033	0.063	0.061	0.038	0.063	0.035	0.066	0.066	0.037
	1 + 8I ($s > 0.3$)	0.088	0.037	0.080	0.045	0.042	0.088	0.035	0.078	0.044	0.037
	1 + 8I ($s > 0.7$)	0.147	0.048	0.142	0.054	0.051	0.143	0.039	0.136	0.044	0.042
	9 – 8I ($s > 0.3$)	0.130	0.046	0.143	0.130	0.048	0.135	0.041	0.145	0.136	0.043
	9 – 8I ($s > 0.7$)	0.079	0.037	0.089	0.066	0.041	0.081	0.037	0.090	0.072	0.040
5 + 20I ($s > 0.5$)	1	0.056	0.038	0.063	0.071	0.041	0.057	0.039	0.064	0.075	0.040
	1 + 8I ($s > 0.3$)	0.084	0.041	0.082	0.051	0.045	0.083	0.038	0.082	0.048	0.039
	1 + 8I ($s > 0.7$)	0.133	0.055	0.137	0.062	0.057	0.129	0.044	0.129	0.049	0.045
	9 – 8I ($s > 0.3$)	0.134	0.052	0.148	0.137	0.054	0.139	0.045	0.154	0.146	0.047
	9 – 8I ($s > 0.7$)	0.081	0.042	0.091	0.074	0.044	0.082	0.043	0.093	0.083	0.043
20	1	0.053	0.046	0.053	0.065	0.047	0.055	0.042	0.055	0.072	0.044
	1 + 8I ($s > 0.3$)	0.086	0.045	0.081	0.050	0.047	0.083	0.043	0.078	0.046	0.044
	1 + 8I ($s > 0.7$)	0.164	0.060	0.160	0.059	0.060	0.163	0.047	0.157	0.048	0.047
	9 – 8I ($s > 0.3$)	0.149	0.059	0.148	0.137	0.060	0.166	0.056	0.164	0.146	0.056
	9 – 8I ($s > 0.7$)	0.086	0.052	0.087	0.079	0.053	0.085	0.051	0.088	0.085	0.052
25 – 5I ($s > 0.5$)	1	0.054	0.044	0.052	0.062	0.047	0.056	0.042	0.056	0.069	0.043
	1 + 8I ($s > 0.3$)	0.087	0.043	0.082	0.047	0.045	0.082	0.042	0.078	0.046	0.043
	1 + 8I ($s > 0.7$)	0.165	0.057	0.160	0.059	0.058	0.163	0.046	0.157	0.045	0.046
	9 – 8I ($s > 0.3$)	0.146	0.058	0.144	0.134	0.059	0.162	0.055	0.158	0.144	0.055
	9 – 8I ($s > 0.7$)	0.084	0.051	0.084	0.075	0.052	0.085	0.049	0.087	0.081	0.049
25 – 20I ($s > 0.5$)	1	0.054	0.035	0.046	0.041	0.038	0.058	0.038	0.049	0.046	0.039
	1 + 8I ($s > 0.3$)	0.087	0.036	0.072	0.037	0.039	0.090	0.038	0.072	0.037	0.039
	1 + 8I ($s > 0.7$)	0.172	0.053	0.158	0.051	0.059	0.171	0.046	0.153	0.042	0.048
	9 – 8I ($s > 0.3$)	0.127	0.054	0.117	0.118	0.053	0.132	0.051	0.118	0.132	0.051
	9 – 8I ($s > 0.7$)	0.074	0.042	0.065	0.060	0.043	0.073	0.047	0.063	0.062	0.045

Notes: See notes under Table 1. The DGP is as in (1) and (2) with $\rho = 1 - c_t T^{-1}$ and $\beta = 0$ and exhibits time-varying variances. For further details see the text.

augmented statistics $\tilde{t}_{ivx}^{\mu_0}$ and $\tilde{t}_{ivx}^{\mu_3}$, the rejection frequencies are between [3.0%, 6.0%] for $T = 200$ and [3.3%, 5.6%] for $T = 500$. Surprisingly the two-sided version of $\tilde{t}_{ivx}^{\mu_2}$ displays in general similar size to $\tilde{t}_{ivx}^{\mu_0}$ and $\tilde{t}_{ivx}^{\mu_3}$, except when an early downward break in the variance occurs ($\sigma_\varepsilon^2(s) = \sigma_v^2(s) = 9 - 8I(s > 0.3)$).¹⁰

5. Real estate returns and rent growth predictability

Price determination in housing markets implies that rents are a fundamental determinant of the housing value and that the rent-to-price ratio (also known as ‘cap rate’) summarizes market expectations of future housing returns or rent growth (cf. Plazzi et al., 2010, Ghysels et al., 2013 and Engsted and Pedersen, 2015 [EP]). The framework of analysis considers the present value relation,

$$r_t - p_t = -\frac{\kappa}{1 - \rho} + E_t \left(\sum_{j=0}^{\infty} \rho^j (h_{t+1+j} - \Delta r_{t+1+j}) \right) \quad (26)$$

where h_t , r_t and p_t are, respectively, the log housing return, the log rent, and the log house price at time t , ρ is the discount factor and κ is a linearization constant; see Campbell et al. (2009), Plazzi et al. (2010) and Ghysels et al. (2013) and EP.

The present value relation in (26) suggests that, holding expected housing returns constant, an increase in expected future rents leads to an increase in today’s house price and originates a decrease in the rent-to-price ratio. Similarly,

¹⁰ As a further robustness check of the findings Tables B.3 and B.4 in the appendix show that, for positive short-run correlation, the other tests do not appear to be sensitive to the sign of the short-run serial correlation of the predictor. The effects are practically the same for both sample sizes, indicating that the size distortions are not finite-sample in nature.

holding the expected rent growth constant, an increase in expected future housing returns must imply a lower price today and thereby an increase of the rent-to-price ratio. Thus, the rent-to-price ratio is a useful measure of valuation if it can predict future housing returns or rent growth.

If expected rent growth and expected housing returns are both stationary, then the rent-to-price ratio should also be stationary, however empirical evidence suggests that this is not always the case; see Kishor and Morley (2015) and the results for $\hat{\rho}$ in the last column of Table 3. Therefore, in what follows we apply the residual-augmented IVX test ($\tilde{t}_{ivx}^{\mu_0}$) and contrast the results obtained with the conventional IVX test (t_{ivx}) of Kostakis et al. (2015), and the conventional OLS based t-ratio computed with and without Newey–West standard errors, t_{OLS} and t_{NW} , respectively.

The analysis in this section contributes to this literature with further evidence on predictability for 19 OECD countries (Australia (AUS), Belgium (BEL), Canada (CAN), Switzerland (CHE), Germany (DEU), Denmark (DNK), Spain (ESP), Finland (FIN), France (FRA) the UK (GBR), Ireland (IRL), Italy (ITA), Korea (KOR), Japan (JPN), the Netherlands (NLD), Norway (NOR), New Zealand (NZL), Portugal (PRT), Sweden (SWE)) and the US. We also look at overall house price indexes for the OECD and the Euro area 16 countries composition (EA16). The data consists of seasonally adjusted quarterly real and nominal house prices, and the rent-to-price ratio from 1970Q1 to 2016Q1 for all countries except Australia (begins 1972Q3), Belgium (begins 1976Q2), Korea (begins 1986Q1), Portugal (begins 1988Q1), Spain (begins 1971Q1), Norway (begins 1979Q1), and Sweden (begins 1980Q1). From the real and nominal house prices, inflation is computed, and from the house prices and the rent-to-price ratio real and nominal rent growth as well as housing returns are obtained. The latter is computed as $H_{t+1} = (P_{t+1} + R_{t+1}) / P_t$.

Table 7 presents the findings for nominal housing returns, nominal rent growth and inflation predictability by the rent-to-price ratio. According to (26) the rent-to-price ratio should predict returns with a positive sign and rent growth with a negative sign. However, regardless of the test used, the results suggest that this is not always the case (which was also noted by EP). The marginal t-tests, t_{OLS} and t_{NW} , suggest that the rent-to-price ratio is a significant predictor in 17 of the 22 countries and economic areas under analysis, whereas $\tilde{t}_{ivx}^{\mu_0}$ and t_{ivx} only find predictability in 12 and 16 of the 22 series, respectively. For instance, t_{NW} finds predictability for ITA, NLD, OECD and EA16, which is not confirmed by both $\tilde{t}_{ivx}^{\mu_0}$ and t_{ivx} . On the other hand, $\tilde{t}_{ivx}^{\mu_0}$ and t_{ivx} find predictability for nominal housing returns in JPN and CHE which was not detected by t_{NW} .

From the results for $\tilde{t}_{ivx}^{\mu_0}$ we also observe that 12 (55%) of the parameter estimates for the 22 returns series are positive ($\hat{\beta}_h > 0$), of which however only eight are statistically significant. The $\tilde{t}_{ivx}^{\mu_0}$ results suggest that of the 12 series for which the rent-to-price ratio has statistically significant predictive power, 4 display a negative $\hat{\beta}_h$. Table 7 shows that there seems to be stronger nominal rent growth predictability than nominal housing return predictability which was also noted in EP. Based on t_{NW} the only cases for which the rent-to-price ratio is not a statistically significant predictor are IRL and SWE. The $\tilde{t}_{ivx}^{\mu_0}$ detects predictability in nominal rent growth for 14 of the series of which six display significant negative parameter estimates ($\hat{\beta}_r < 0$).

Overall, the number of significant statistics is greater for t_{NW} (and t_{OLS}) than for t_{ivx} and $\tilde{t}_{ivx}^{\mu_0}$ when nominal housing returns and rent growth are considered. In contrast to t_{NW} , $\tilde{t}_{ivx}^{\mu_0}$ does not find significant results for DNK, FRA, ITA, NLD, GBR, OECD and EA16 when housing returns are considered and for CAN, FIN, FRA, GBR, the US, OECD and EA16 when the rent growth is analyzed. However, $\tilde{t}_{ivx}^{\mu_0}$ finds significant results for JPN and CHE for housing returns, and for SWE for rent growth, whereas t_{NW} (and t_{OLS}) does not.

When real housing returns are used,¹¹ $\tilde{t}_{ivx}^{\mu_0}$ is only significant for BEL and KOR (whereas when nominal housing returns were used evidence was found for 12 countries), and when real rent growth is considered for BEL, CAN, FIN, KOR, NLD, ESP and GBR (whereas for nominal rent growth 14 countries displayed evidence of predictability). According to EP, the present value relation in (26) shows that if the rent-to-price ratio does not predict future housing returns, then it must predict future rent growth. In other words, since the rent-to-price ratio varies over time then either expected housing returns or expected rent growth, or both, must also vary over time, i.e., the null hypothesis cannot consist of having unpredictable housing returns and unpredictable rent growth.

Defining log inflation from t to $t + 1$ as π_{t+1} , and replacing h_{t+1+j} and Δr_{t+1+j} in (26) by $(h_{t+1+j} - \pi_{t+1+j})$ and $(\Delta r_{t+1+j} - \pi_{t+1+j})$, respectively, the resulting representation suggests that if the rent-to-price ratio predicts nominal and real housing returns or rent growth differently, then it must be due to the rent-to-price ratio having predictive power for future inflation. For example, if nominal housing returns are predictable but inflation is not, then the result must be that real housing returns are predictable. The evidence of predictability found with the nominal returns for DEU, JPN, NZL, PRT and CHE is not observed when real returns are used, but significant results are observed when analyzing predictability of inflation for these countries. From (26) it is clear that a negative return coefficient using nominal data can only turn positive if the inflation coefficient is negative and numerically larger than the return coefficient. Note that the rent-to-price ratio predicts inflation with a negative sign for DEU, JPN, PRT and CHE, which displayed negative return coefficients when nominal data was used but these are not significant for real housing returns.

From the results of the predictability tests performed we conclude, using the four competing estimators that five countries actually emerge for which the rent-to-price ratio is a significant predictor for housing returns and rent growth in nominal terms. To be specific:

¹¹ Results for real housing returns can be obtained from the authors.

Table 7

Predictability of nominal returns, nominal rent growth and inflation. Sample: 1970–2016.

	House price returns					Rent growth					Inflation					
	$\tilde{t}_{IVX}^{\mu_0}$	t_{IVX}	t_{OLS}	t_{NW}	$\frac{\hat{\sigma}_{uv}}{\hat{\sigma}_u \hat{\sigma}_v}$	$\tilde{t}_{IVX}^{\mu_0}$	t_{IVX}	t_{OLS}	t_{NW}	$\frac{\hat{\sigma}_{uv}}{\hat{\sigma}_u \hat{\sigma}_v}$	$\tilde{t}_{IVX}^{\mu_0}$	t_{IVX}	t_{OLS}	t_{NW}	$\frac{\hat{\sigma}_{uv}}{\hat{\sigma}_u \hat{\sigma}_v}$	$\tilde{\rho}_{cap}$
BEL	3.93***	3.23***	5.66***	4.45***	−0.70	5.36***	4.22***	7.51***	6.50***	0.19	1.05	0.61	1.06	0.81	0.05	0.999
	4.05***	3.49***	4.25***	4.33***	−0.79	10.46***	6.15***	5.68***	6.98***	0.39	1.53	1.87*	1.65*	1.02	0.13	0.997
CAN	2.05**	1.78*	6.11***	5.29***	−0.83	1.46	0.93	7.16***	4.58***	0.22	0.60	0.25	4.20***	2.89***	0.15	0.994
DNK	1.45	1.95*	3.26***	2.20**	−0.77	2.80***	2.11**	4.01***	3.52***	0.12	1.24	0.83	1.64	1.10	0.12	0.989
FIN	1.87*	2.11**	4.67***	4.51***	−0.44	1.43	0.96	2.93***	2.44**	0.50	1.34	1.09	3.13***	2.27**	0.15	0.991
FRA	1.38	1.69*	5.08***	3.57***	−0.37	0.89	0.63	3.92***	3.65***	0.20	−0.58	−0.72	1.12	0.81	0.19	0.997
DEU	−3.99***	−3.09***	−2.71**	−1.97**	−0.41	−6.90***	−4.99***	−7.44***	−10.45***	0.66	−6.13***	−5.42***	−7.24***	−5.54***	0.03	0.996
IRL	2.08**	2.59***	7.99***	6.94***	−0.43	−0.23	−0.52	0.17	0.14	0.69	−0.78	−1.10	0.86	0.60	0.04	0.988
ITA	−0.61	−0.61	3.27***	1.97**	−0.53	−2.26**	−3.10***	−1.61	−1.74*	0.37	−3.87***	−4.15***	−7.03***	−7.21***	−0.03	0.965
JPN	−3.03***	−2.88***	−0.82	−0.82	−0.50	−9.57***	−7.63***	−5.39***	−6.17***	−0.07	−8.02***	−8.23***	−6.17***	−5.72***	0.07	0.996
KOR	−1.26	−1.60	−1.05	−0.75	−0.68	−4.65***	−6.17***	−6.72***	−3.73***	0.06	1.59	1.71*	1.30	0.92	0.34	0.985
NLD	1.53	2.31**	4.01***	4.42***	−0.54	1.77*	1.09	1.82**	2.23**	0.44	−2.39**	−2.42**	−1.73*	−1.31	0.00	0.988
NZL	3.80***	3.05***	5.35***	4.54***	−0.37	4.45***	3.07***	5.54***	7.21***	0.42	3.09***	2.21**	3.51***	2.84***	0.13	0.999
NOR	2.60***	1.85*	3.13***	2.97***	−0.73	5.25***	2.85***	4.88***	3.54**	0.22	1.06	0.94	1.88*	1.73*	0.15	0.996
PRT	−1.66*	−1.77*	−3.66***	−2.43**	−0.47	−4.97***	−2.93***	−6.19***	−3.70***	0.34	−4.96***	−5.96***	−11.44***	−9.78***	0.12	0.992
ESP	2.54**	2.72***	10.17***	7.41***	−0.54	2.28**	2.26**	11.95***	10.00***	0.18	−1.48	−1.82*	−0.83	−0.64	0.08	0.994
SWE	−0.28	−0.39	0.21	0.19	−0.45	1.91*	1.54	−0.50	−0.76	0.44	2.34**	2.13**	0.01	−0.11	0.23	0.995
CHE	−2.76***	−2.19**	−1.49	−1.25	−0.78	−8.83***	−9.66***	−10.03***	−7.50***	0.30	−9.14***	−8.74***	−9.45***	−7.20***	0.19	0.977
GBR	1.57	1.85*	4.18***	3.75***	−0.43	1.47	0.80	2.82***	3.44***	0.56	0.20	−0.14	0.43	0.27	0.26	0.990
USA	−1.38	−0.79	−0.08	−0.04	−0.37	−1.43	−2.82***	−2.80***	−1.97**	0.45	−3.60***	−3.46***	−3.84***	−4.14***	0.21	0.977
OECD	−0.58	−0.10	5.50***	4.09***	−0.31	−1.22	−1.53	3.42***	3.90***	0.30	−2.27**	−2.83***	−1.75*	−1.66*	0.27	0.988
EA16	−0.20	0.11	5.71***	4.01***	−0.40	−1.42	−1.74*	3.05***	3.04***	0.31	−2.52* [*]	−3.13***	−2.16**	−1.84*	0.12	0.986

Note: $\tilde{\rho}_{cap}$ corresponds to the sum of the autoregressive parameters of an estimated AR model for the predictor (using AIC as a lag selection criteria) and $\hat{\sigma}_{uv}/(\hat{\sigma}_u \hat{\sigma}_v)$ is the correlation between the residuals of each of the predictive regressions considered (the house price returns, rent growth and inflation) and the residuals of the AR model of the predictor. t_{IVX} refers to the conventional IVX test, $\tilde{t}_{IVX}^{\mu_0}$ to the residual augmented IVX test, and t_{OLS} and t_{NW} , to the conventional OLS based t-ratio computed with and without Newey–West standard errors, respectively. House price returns are computed as earlier indicated.

*Significance at the 10%.

**Significance at the 5%.

***Significance at the 1%.

- (a) The findings reveal that the rent-to-price ratio (in nominal terms) is a dominant and stable predictor for both future housing returns and rent growth for Australia, Germany, Portugal, Spain and Japan.
- (b) Of these five countries where the rent-to-price ratio is found to be stable in terms of predictive power and the sign of the coefficient, three countries (Germany, Japan and Portugal) are shown to have a negative predictive relationship with future housing returns. That is, using robust inference methods, the pattern that stands out is that rent-to-price ratios negatively predict housing returns: a divergence from the dynamic Gordon Growth model, as stated by EP (2015).
- (c) Interestingly, rent growth in Belgium turns out to be the most predictable variable of all. All four competing estimators predict future rent in both nominal and real terms. Rent growth is also consistently predictable in Spain and Korea, but the predictive coefficient of rent growth in Spain changes signs when shifting the analysis from nominal to real terms.

6. Concluding remarks

This paper introduced a new IVX-based statistic computed from a residual augmented predictive regression motivated by Amihud and Hurvich (2004), and reexamined the empirical evidence on house price returns and rent growth predictability using these new robust methods.

The residual-augmented IVX variant allows practitioners to distinguish more reliably between the null of no predictability and the alternative. The method is asymptotically correct under near-integration as well as under stationarity of the regressor, has improved local power under high regressor persistence, and allows, e.g., for heterogeneity of the data in the form of time-varying variances. Moreover, the residual-augmented IVX procedure also displays considerably better size control of one-sided tests (right and left tailed tests) than the original IVX. This is an important feature, since many empirical predictive regressions are run with one-sided tests.

The results derived here on bias correction can be generalized for other types of instrumental variable estimation than just IVX. The IV framework of Breitung and Demetrescu (2015), which distinguishes between type-I instruments that are less persistent than the initial regressor (the IVX instrument is actually of type I), and type-II instruments that are (stochastically) trending, yet exogenous, allows for a quick discussion: a careful examination of the arguments presented here shows that they are easily extended for type-I instruments, but type-II instruments behave like the OLS estimator where residual-augmentation is not improving on the test procedure asymptotically.

The provided Monte Carlo evidence shows that the asymptotic improvements translate, all in all, into good finite-sample performance, also in the presence of time-varying volatility or time varying persistence. Although the demeaning scheme in the autoregression for the predictor does not matter asymptotically, it does so in samples of usual sizes and it is recommended to resort to recursive demeaning or subtracting the initial condition to account for nonzero predictor means when computing the required residuals.

Finally, the analysis of OECD housing price data showed that the bias-adjusted IVX procedure detected predictability more often than the original IVX procedure, but less often than non-robust procedures. Overall, this analysis reveals, among other things, that the rent-to-price ratio (in nominal terms) is a useful predictor for both future housing returns and rent growth for Australia, Germany, Portugal, Spain and Japan. We leave to further work the check of whether adding other putative predictors (such as, among others, disposable income, mortgage rates, unemployment, investment in housing and short-term interest rates) strengthens the evidence on predictability.

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Appendix A. Preliminary results

Throughout the proofs we consider that $\sum_{j=0}^{t-1} \varrho^{kj} = \frac{1-\varrho^{kt}}{1-\varrho^k} = \frac{T^\eta}{a} \left(\frac{1-\varrho^{kt}}{1+\varrho+\dots+\varrho^{k-1}} \right) \leq \frac{1}{ka} T^\eta$ for large enough T and fixed k , where $\varrho := 1 - \frac{a}{T^\eta}$ with $\eta \in (0, 1)$ and $a > 0$ and fixed. Furthermore, let C denote a generic constant whose value may change from occurrence to occurrence.

Lemma A.1. Under the assumptions of Theorem 3.1, as $T \rightarrow \infty$, it follows that

1. $\frac{1}{T} \sum_{t=p+1}^T \ddot{\mathbf{x}}_{t-1} \ddot{\mathbf{x}}_{t-p}' \xrightarrow{p} \boldsymbol{\alpha}'_p \int_0^1 \sigma_v^2(s) ds$, where $\boldsymbol{\alpha}_p := (\alpha_0, \dots, \alpha_{p-1})$, $\ddot{\mathbf{x}}_{t-p} := (\ddot{x}_{t-1}, \dots, \ddot{x}_{t-p})'$ and α_h is as defined in Theorem 3.1;
2. $\frac{1}{T} \sum_{t=p+1}^T \ddot{\mathbf{x}}_{t-p} \ddot{\mathbf{x}}_{t-p}' \xrightarrow{p} \boldsymbol{\Omega} \int_0^1 \sigma_v^2(s) ds$, where $\boldsymbol{\Omega}$ is a $p \times p$ matrix with generic element $a_{ij} = \alpha_{|i-j|}$;
3. $\frac{1}{T} \sum_{t=p+1}^T \ddot{\mathbf{x}}_{t-p} \ddot{\mathbf{x}}_{t-p}' v_t^2 \xrightarrow{p} \boldsymbol{\Omega} \int_0^1 \sigma_v^4(s) ds$;
4. $\frac{1}{T} \sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2 \xrightarrow{p} \alpha_0 \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds$.

Proof of Lemma A.1. Lemma 1 in Phillips and Xu (2006) can immediately be used to conclude that $T^{-1} \sum_{t=h+1}^T \ddot{x}_t \ddot{x}_{t-h} \xrightarrow{p} \alpha_h \int_0^1 \sigma_v^2(s) ds$, $h = 0, 1, \dots, p-1$; this suffices to establish the results in the first two items. The result in item 3 also follows directly from Lemma 1 of Phillips and Xu (2006), and the proof can be adapted in a straightforward manner to establish the result in item 4. ■

Lemma A.2. Under the assumptions of Theorem 3.2, as $T \rightarrow \infty$, it follows that

$$\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{-\frac{1}{2}} \sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t \xrightarrow{d} \mathcal{N}(0, 1)$$

where $\tilde{z}_t := \sum_{j=0}^{t-1} \varrho^j v_{t-j}$.

Proof of Lemma A.2. Consider $s_T^2 := \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sigma_{v,t-1-j}^2 \sigma_{\varepsilon,t}^2$ and note that s_T^2 is bounded and bounded away from zero, since

$$\frac{\min_{1 \leq t \leq T} \sigma_{v,t}^2 \min_{1 \leq t \leq T} \sigma_{\varepsilon,t}^2}{T^{1+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \leq s_T^2 \leq \frac{\max_{1 \leq t \leq T} \sigma_{v,t}^2 \max_{1 \leq t \leq T} \sigma_{\varepsilon,t}^2}{T^{1+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j}$$

where $\sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sim CT^{1+\eta}$.

Since,

$$\frac{\sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{\frac{1}{2}}} = \frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T \frac{\tilde{z}_{t-1} \varepsilon_t}{s_T} \left(\frac{\sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sigma_{v,t-1-j}^2 \sigma_{\varepsilon,t}^2}{\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2} \right)^{\frac{1}{2}} \quad (\text{A.1})$$

we show next that $\frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T \frac{\tilde{z}_{t-1} \varepsilon_t}{s_T}$ follows a limiting standard normal distribution by resorting to a central limit theorem for martingale difference [md] arrays (Davidson, 1994, Theorem 24.3). However, to apply it, we need to show that, (i) $\max_t \frac{1}{T^{1/2+\eta/2}} \left| \frac{\tilde{z}_{t-1} \varepsilon_t}{s_T} \right| \xrightarrow{p} 0$ and (ii) $\frac{1}{T^{1+\eta}} \sum_{t=2}^T \frac{\tilde{z}_{t-1}^2 \varepsilon_t^2}{s_T^2} \xrightarrow{p} 1$.

Given that the result in (ii) also implies

$$\left(\frac{\sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sigma_{v,t-1-j}^2 \sigma_{\varepsilon,t}^2}{\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2} \right)^{\frac{1}{2}} \xrightarrow{p} 1, \quad (\text{A.2})$$

hence the result in (A.1) will follow.

To verify (i) note that uniform boundedness of moments of order $2 + \delta^*$ for some $\delta^* > 0$ of $T^{-\eta/2} \tilde{z}_{t-1} \varepsilon_t$ suffices to establish this condition. An application of Hölder's inequality shows that uniformly bounded 4th order moments of $T^{-\eta/2} \tilde{z}_{t-1}$ and uniform $L_{4+\delta^*}$ -boundedness of ε_t suffices, since δ^* may be chosen arbitrarily close to zero, so we check the uniform boundedness of

$$\mathbb{E} \left(\frac{\tilde{z}_{t-1}^4}{T^{2\eta}} \right) = \frac{1}{T^{2\eta}} \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{t-2} \sum_{m=0}^{t-2} \varrho^j \varrho^k \varrho^l \varrho^m \mathbb{E} (v_{t-j} v_{t-k} v_{t-l} v_{t-m}). \quad (\text{A.3})$$

Due to the serial independence of v_t , the expectation $\mathbb{E} (v_{t-j} v_{t-k} v_{t-l} v_{t-m})$ is nonzero only if the indices are pairwise equal, thus we can simplify (A.3) as,

$$\mathbb{E} \left(\frac{\tilde{z}_{t-1}^4}{T^{2\eta}} \right) = \frac{1}{T^{2\eta}} \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^{2j} \varrho^{2k} \mathbb{E} (v_{t-j}^2 v_{t-k}^2).$$

Since v_t is uniformly L_4 -bounded, the expectations on the r.h.s. are uniformly bounded for any t, k and j , therefore,

$$0 \leq E \left(\frac{\tilde{z}_{t-1}^4}{T^{2\eta}} \right) \leq C \frac{1}{T^{2\eta}} \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^{2j} \varrho^{2k} = C \frac{1}{T^{2\eta}} \left(\sum_{j=0}^{t-2} \varrho^{2j} \right)^2 \leq C \frac{1}{T^{2\eta}} \left(\sum_{j=0}^{T-2} \varrho^{2j} \right)^2 \leq C$$

which suffices for the required uniform L_4 -boundedness.

To check condition (ii), it suffices to show that

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 - s_T^2 \xrightarrow{p} 0 \quad (\text{A.4})$$

because s_T^2 is bounded and bounded away from zero (we learn from [Lemma A.4](#) that $s_T^2 \rightarrow \frac{1}{2a} \int_0^1 \sigma_v^2(s) \sigma_u^2(s) ds$, but the exact limit does not matter here). To prove (A.4), write

$$\begin{aligned} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 &= \sum_{t=2}^T \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} (\varepsilon_t^2 - \sigma_{\varepsilon,t}^2) + \sum_{t=2}^T \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \sigma_{\varepsilon,t}^2 \\ &=: A_T + B_T. \end{aligned}$$

Note that $\sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} (\varepsilon_t^2 - \sigma_{\varepsilon,t}^2)$ builds an md array and as such, is uncorrelated in t . Hence, showing $\frac{1}{T^{1+\eta}} A_T$ to vanish is not difficult, given that from the uncorrelatedness of the summands we can write that,

$$\begin{aligned} \text{Var} \left(\frac{1}{T^{1+\eta}} A_T \right) &= \frac{1}{T^{2+2\eta}} \sum_{t=2}^T \text{Var} \left[\sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} (\varepsilon_t^2 - \sigma_{\varepsilon,t}^2) \right] \\ &= \frac{1}{T^{2+2\eta}} \sum_{t=2}^T E \left(\sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \right)^2 E (\varepsilon_t^2 - \sigma_{\varepsilon,t}^2)^2. \end{aligned}$$

Now, ε_t is uniformly L_4 -bounded and

$$E \left(\sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \right)^2 = \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{t-2} \sum_{m=0}^{t-2} \varrho^j \varrho^k \varrho^l \varrho^m E (v_{t-1-j} v_{t-1-k} v_{t-1-l} v_{t-1-m})$$

where the expectation on the r.h.s. is, as before, uniformly bounded and nonzero only if the indices are pairwise equal. Hence,

$$0 \leq E \left(\sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \right)^2 \leq C \sum_{j=0}^{t-2} \sum_{k=0}^{t-2} \varrho^{2j} \varrho^{2k} \leq CT^{2\eta}$$

leading to $\text{Var} (T^{-1-\eta} A_T) \rightarrow 0$ and thus $A_T = o_p (T^{1+\eta})$.

Regarding B_T , note that,

$$\begin{aligned} B_T &= T^{1+\eta} s_T^2 + \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} (v_{t-1-j}^2 - \sigma_{v,t-1-j}^2) \sigma_{\varepsilon,t}^2 + \sum_{t=2}^T \sum_{j=0}^{t-2} \sum_{\substack{k=0 \\ j \neq k}}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \sigma_{\varepsilon,t}^2 \\ &= T^{1+\eta} s_T^2 + B_{T1} + B_{T2}. \end{aligned}$$

For B_{T1} we have from the serial independence and L_4 -boundedness of v_t that

$$\begin{aligned} E \left(\sum_{j=0}^{t-2} \varrho^{2j} (v_{t-1-j}^2 - \sigma_{v,t-1-j}^2) \sigma_{\varepsilon,t}^2 \right)^2 &= \sigma_{\varepsilon,t}^4 \sum_{j=0}^{t-2} \varrho^{4j} E (v_{t-1-j}^2 - \sigma_{v,t-1-j}^2)^2 \\ &\leq CT^\eta \end{aligned}$$

and thus $E \left| \sum_{j=0}^{t-2} \varrho^{2j} (v_{t-1-j}^2 - \sigma_{v,t-1-j}^2) \sigma_{\varepsilon,t}^2 \right| \leq CT^{\eta/2}$. Hence, $E |T^{-1-\eta} B_{T1}| \leq CT^{-1-\eta} \sum_{t=2}^T T^{\eta/2} \rightarrow 0$ and Markov's inequality indicates that $B_{T1} = o_p (T^{1+\eta})$.

For B_{T2} we proceed similarly,

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{t=2}^T \sum_{j=0}^{t-2} \sum_{\substack{k=0 \\ j \neq k}}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \sigma_{\varepsilon,t}^2 \right)^2 \right] \\ = \sum_{t=2}^T \sum_{s=2}^T \sum_{j=0}^{t-2} \sum_{\substack{k=0 \\ j \neq k}}^{t-2} \sum_{l=0}^{s-2} \sum_{\substack{m=0 \\ l \neq m}}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m \sigma_{\varepsilon,t}^2 \sigma_{\varepsilon,s}^2 \mathbb{E} (v_{t-1-j} v_{t-1-k} v_{s-1-l} v_{s-1-m}), \end{aligned}$$

where the expectations on the r.h.s. are nonzero if $t-j = s-l$ and $t-k = s-m$ or if $t-j = s-m$ and $t-k = s-l$ (with $t-j = t-k$ and $s-l = s-m$ being excluded by the requirement that $j \neq k$ and $l \neq m$). Note that, for any t, s, j, k, l, m with $j \neq k$ and $l \neq m$,

$$\sigma_{\varepsilon,t}^2 \sigma_{\varepsilon,s}^2 \mathbb{E} (v_{t-1-j} v_{t-1-k} v_{s-1-l} v_{s-1-m}) \leq \left(\max_t \sigma_{\varepsilon,t}^2 \right)^2 \left(\max_t \sigma_{\varepsilon,t}^2 \right)^2 \leq C.$$

Let us now focus on the terms for which $t-s = j-l = k-m$. Thus, for $t = s, t = 2, \dots, T$, we obtain

$$\sum_{\substack{j=0 \\ j \neq k, l \neq m, t-s=j-l=k-m}}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{s-2} \sum_{m=0}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m = \sum_{j=0}^{t-2} \sum_{\substack{k=0 \\ j \neq k}}^{t-2} \varrho^{2j} \varrho^{2k} \leq \left(\sum_{j=0}^{t-2} \varrho^{2j} \right)^2;$$

and for $s = t-1, t = 3, \dots, T$, we have analogously that,

$$\sum_{\substack{j=0 \\ j \neq k, l \neq m, t-s=j-l=k-m}}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{s-2} \sum_{m=0}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m \leq \varrho^2 \left(\sum_{j=0}^{t-3} \varrho^{2j} \right)^2$$

while for $s = t+1, t = 2, \dots, T-1$ (or equivalently $t = s-1, s = 3, \dots, T$) it follows that,

$$\sum_{\substack{j=0 \\ j \neq k, l \neq m, t-s=j-l=k-m}}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{s-2} \sum_{m=0}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m \leq \varrho^2 \left(\sum_{l=0}^{s-3} \varrho^{2l} \right)^2.$$

Repeating the discussion for $s = t \pm r$ for $r = 2, \dots, T-2$, we have

$$\sum_{\substack{j=0 \\ j \neq k, l \neq m, t-s=j-l=k-m}}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{s-2} \sum_{m=0}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m \leq 2\varrho^{2r} \left(\sum_{j=0}^{t-r-2} \varrho^{2j} \right)^2,$$

leading to

$$\sum_{t=2}^T \sum_{s=2}^T \sum_{\substack{j=0 \\ j \neq k, l \neq m, t-s=j-l=k-m}}^{t-2} \sum_{k=0}^{t-2} \sum_{l=0}^{s-2} \sum_{m=0}^{s-2} \varrho^j \varrho^k \varrho^l \varrho^m \leq \sum_{t=2}^T \left(\sum_{j=0}^{t-2} \varrho^{2j} \right)^2 + 2 \sum_{r=1}^{T-2} \varrho^{2r} \sum_{t=2+r}^T \left(\sum_{j=0}^{t-r-2} \varrho^{2j} \right)^2.$$

The same holds when imposing $t-s = j-m = k-l$, such that, with $\sum_{j=0}^{t-r-2} \varrho^{2j} \leq \sum_{j=0}^{t-1} \varrho^{2j}$ and $\sum_{t=2+r}^T C \leq CT$, thus, we ultimately have

$$\mathbb{E} \left(\sum_{t=2}^T \sum_{j=0}^{t-2} \sum_{\substack{k=0 \\ j \neq k}}^{t-2} \varrho^j \varrho^k v_{t-1-j} v_{t-1-k} \sigma_{\varepsilon,t}^2 \right)^2 \leq CT^{1+3\eta}$$

and consequently $B_{T2} = o_p(T^{1+\eta})$ when $\eta < 1$, as required to complete the proof. ■

Lemma A.3. Under the assumptions of Theorem 3.2, it follows, as $T \rightarrow \infty$, that

- (i) $\left(\sum_{t=2}^T z_{t-1}^2 \varepsilon_t^2 \right)^{-1/2} \sum_{t=2}^T z_{t-1} \varepsilon_t \xrightarrow{d} \mathcal{N}(0, 1)$; and
- (ii) $\left(\sum_{t=2}^T z_{t-1}^2 u_t^2 \right)^{-1/2} \sum_{t=2}^T z_{t-1} u_t \xrightarrow{d} \mathcal{N}(0, 1)$.

Lemma A.3 suggests the use of Eicker–White standard errors in the heteroskedastic near-integrated case, $W.s.e := \frac{(\sum_{t=2}^T z_{t-1}^2 \hat{\varepsilon}_t^2)^{1/2}}{\sum_{t=2}^T z_{t-1}^2}$ with $\hat{\varepsilon}_t$ the OLS residuals guaranteeing $\sup_{2 \leq t \leq T} |\hat{\varepsilon}_t - \varepsilon_t| \xrightarrow{P} 0$ both in cases with and without intercept, and also better finite-sample behavior; see [Kostakis et al. \(2015\)](#). For the stable case, Eicker–White standard errors are “mandatory” under time heteroskedasticity ([Phillips and Xu, 2006](#)).

Proof of Lemma A.3. We first resort to the Phillips–Solo decomposition of v_t and write $v_t = \omega v_t + \Delta \tilde{v}_t$ where $\tilde{v}_t$ is a linear process in v_t with exponentially decaying coefficients. Let also $\tilde{z}_t := (1 - \varrho L)_+^{-1} v_t$. Thus, denoting $\tilde{z}_t := \sum_{j=0}^{t-1} \varrho^j v_{t-j}$ like in [Lemma A.2](#), it follows that,

$$\begin{aligned} \tilde{z}_t &= \omega \sum_{j=0}^{t-1} \varrho^j v_{t-j} + \left(\tilde{v}_t + (\varrho - 1) \sum_{j=1}^{t-1} \varrho^{j-1} \tilde{v}_{t-j} - \varrho^{t-1} \tilde{v}_1 \right) \\ &= \omega \tilde{z}_t + d_t, \end{aligned}$$

and it can then easily be shown that $\text{Var} \left(\sum_{j=1}^{t-1} \varrho^{j-1} \tilde{v}_{t-j} \right) \leq CT^\eta$ such that d_t is uniformly L_2 -bounded given that $\varrho - 1 = -aT^{-\eta}$. Similarly, $T^{-\eta/2} \tilde{z}_t$ is uniformly L_2 -bounded itself. We now show that

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 = \frac{\omega^2}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 + o_p(1) \quad (\text{A.5})$$

and

$$\frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t = \frac{\omega}{T^{1/2+\eta/2}} \sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t + o_p(1). \quad (\text{A.6})$$

Let us consider first (A.5). Note that,

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 = \frac{\omega^2}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 + \frac{2\omega}{T^{1+\eta}} \sum_{t=2}^T \tilde{z}_{t-1} d_{t-1} \varepsilon_t^2 + \frac{1}{T^{1+\eta}} \sum_{t=2}^T d_{t-1}^2 \varepsilon_t^2.$$

Since, $E(|d_{t-1}^2 \varepsilon_t^2|) = E(d_{t-1}^2) E(\varepsilon_t^2)$ and $E(|\tilde{z}_{t-1} d_{t-1} \varepsilon_t^2|) \leq (E(\tilde{z}_{t-1}^2) E(d_{t-1}^2))^{1/2} E(\varepsilon_t^2)$ due to the independence of ε_t and d_{t-1} and of ε_t and z_{t-1} . With $E(d_{t-1}^2)$, $E(\varepsilon_t^2)$ and $T^{-\eta} E(\tilde{z}_{t-1}^2)$ being uniformly bounded, (A.5) then follows. To establish (A.6), write

$$\frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t = \frac{\omega}{T^{1/2+\eta/2}} \sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t + \frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T d_{t-1} \varepsilon_t$$

and note that $d_{t-1} \varepsilon_t$ has the md property. Hence, $\sum_{t=2}^T d_{t-1} \varepsilon_t = o_p(T^{1/2})$ due to the uniform L_2 -boundedness and independence of ε_t and d_{t-1} . Thus, from (A.5) and (A.6) we obtain that

$$\frac{\sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{1/2}} - \frac{\sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{1/2}} \xrightarrow{P} 0. \quad (\text{A.7})$$

In a second step we use the same reasoning to show that

$$\frac{\sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{1/2}} - \frac{\sum_{t=2}^T z_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T z_{t-1}^2 \varepsilon_t^2 \right)^{1/2}} \xrightarrow{P} 0. \quad (\text{A.8})$$

Write to this end $z_t := \tilde{z}_t + r_t$ where $r_t := -(1 - \varrho L)_+^{-1} \frac{c_t}{T} x_{t-1}$ with

$$\text{Var} \left(\frac{1}{\sqrt{T}} x_t \right) = \frac{1}{T} \sum_{j=1}^t \sum_{k=1}^t \left(1 - \frac{c_{t-j}}{T} \right)^j \left(1 - \frac{c_{t-k}}{T} \right)^k E(v_{t-j} v_{t-k}) \leq \frac{1}{T} \sum_{j=1}^t \sum_{k=1}^t |E(v_{t-j} v_{t-k})|.$$

Given the uniform L_2 -boundedness of the innovations v_t and the exponential decay of the Wold coefficients of v_t , $|E(v_{t-j} v_{t-k})| \leq C e^{b|j-k|} \forall t$ and $\frac{1}{\sqrt{T}} x_t$ is easily shown to be uniformly L_2 -bounded.

The key in establishing (A.8) is to note that r_{t-1} is independent of ε_t and uniformly L_2 -bounded, and that $T^{-\eta} E(z_{t-1}^2)$ is uniformly bounded too whenever $T^{-\eta} E(\tilde{z}_{t-1}^2)$ and $E(r_t^2)$ are. The arguments employed to show (A.7) thus apply for z_t and $\tilde{z}_t$ as well, and (A.8) holds.

Summing up, $\frac{\sum_{t=2}^T z_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T z_{t-1}^2 \varepsilon_t^2 \right)^{1/2}}$ and $\frac{\sum_{t=2}^T \tilde{z}_{t-1} \varepsilon_t}{\left(\sum_{t=2}^T \tilde{z}_{t-1}^2 \varepsilon_t^2 \right)^{1/2}}$ are asymptotically equivalent and the result follows from [Lemma A.2](#).

The proof of the result in (ii) follows along the same lines and we omit the details. ■

Lemma A.4. Under the assumptions of [Theorem 3.2](#), it holds, as $T \rightarrow \infty$, that

1. $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2 \xrightarrow{P} \frac{\omega^2}{2a} \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds$;
2. $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1}^2 u_t^2 \xrightarrow{P} \frac{\omega^2}{2a} \int_0^1 \sigma_v^2(s) \sigma_u^2(s) ds$ where $\sigma_u^2(s) = \sigma_\varepsilon^2(s) + \gamma^2 \sigma_v^2(s)$;
3. $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} \ddot{X}_{t-1} \Rightarrow \frac{\omega^2}{a} \left(X(1) \ddot{X}(1) - \int_0^1 X(s) dM(s) + \int_0^1 c(s) X^2(s) ds \right)$ where $\ddot{X}(s) = X(s) - \int_0^1 X(r) dr$ and $X(r)$ is an Ornstein–Uhlenbeck type process as defined in [\(15\)](#).

Proof of Lemma A.4. 1. To obtain the limit of $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2$, we use from the proof of [Lemma A.3](#) (see [\(A.2\)](#)) the fact that

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2 = \omega^2 \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sigma_{v,t-1-j}^2 \sigma_{\varepsilon,t}^2 + o_p(1).$$

The Lipschitz property implies that $|\sigma_{v,t-1-j}^2 - \sigma_{v,t}^2| \leq C \frac{j}{T}$ such that

$$0 \leq \frac{1}{T^{1+\eta}} \left| \sum_{t=2}^T \sum_{j=0}^{t-2} \varrho^{2j} \sigma_{v,t-1-j}^2 \sigma_{\varepsilon,t}^2 - \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \sum_{j=0}^{t-2} \varrho^{2j} \right| \leq C \frac{1}{T^{2+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} j \varrho^{2j}.$$

On the r.h.s. we have immediately, as $T \rightarrow \infty$, that

$$\frac{1}{T^{2+\eta}} \sum_{t=2}^T \sum_{j=0}^{t-2} j \varrho^{2j} \rightarrow 0$$

given that $\sum_{j=0}^{t-2} j \varrho^{2j} = \frac{t \varrho^{2(t-3)} (\varrho - 1) - (\varrho^{2(t-2)} - 1)}{(\varrho^2 - 1)^2}$, where $\left| \frac{t \varrho^{2(t-3)} (\varrho - 1)}{(\varrho^2 - 1)^2} \right| \leq CT^{1+\eta} \varrho^{2(t-3)}$ and $\left| \frac{\varrho^{2(t-2)} - 1}{(\varrho^2 - 1)^2} \right| \leq CT^{2\eta}$. We also observe that,

$$\begin{aligned} \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \sum_{j=0}^{t-2} \varrho^{2j} &= \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \frac{T^\eta}{a} \left(\frac{1 - \varrho^{2(t-1)}}{1 + \varrho} \right) \\ &= \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \frac{T^\eta}{a(1 + \varrho)} - \frac{1}{T^{1+\eta}} \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \frac{T^\eta}{a} \left(\frac{\varrho^{2(t-1)}}{1 + \varrho} \right). \end{aligned}$$

The first summand on the r.h.s. is easily seen to converge to $\frac{1}{2a} \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds$, while, for the second, we have

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T \sigma_{v,t}^2 \sigma_{\varepsilon,t}^2 \frac{T^\eta}{a} \left(\frac{\varrho^{2(t-1)}}{1 + \varrho} \right) \leq \frac{C}{aT} \sum_{t=2}^T \varrho^{2(t-1)} = O(T^{\eta-1}) = o(1)$$

as required to complete the proof.

2. The proof of 2 is analogous to the proof of 1 and is therefore omitted.

3. Let $S_t := \sum_{j=2}^t Z_t$. We first follow [Breitung and Demetrescu \(2015, Proof of Corollary 1.2\)](#) and show that

$$\frac{1}{T^{1/2+\eta}} S_t = \frac{1}{aT^{1/2}} x_t + R_{t,T}$$

where $\left[E(|R_{t,T}|^2) \right]^{1/2} \rightarrow 0$ as $T \rightarrow \infty$ uniformly in $t = 1, \dots, T$. The arguments are essentially the same as there; the only difference is having to show that $\left[E(|x_t - x_{t-j}|^2) \right]^{1/2} \leq Cj^{1/2}$ for all t and j , which is obvious in their strictly stationary setup, but marginally more difficult here. To this end, recall that $\Delta x_t := v_t - \frac{c_{t-1}}{T} x_{t-1}$ and use Minkowski's inequality to conclude that,

$$\begin{aligned} \left[E((x_t - x_{t-j})^2) \right]^{1/2} &= \left[E \left(\left(\sum_{k=0}^{j-1} v_{t-j-k} - \frac{1}{T} \sum_{k=0}^{j-1} c_{t-k-1} x_{t-k-1} \right)^2 \right) \right]^{1/2} \\ &\leq \left[E \left(\left(\sum_{k=0}^{j-1} v_{t-j-k} \right)^2 \right) \right]^{1/2} + \frac{1}{\sqrt{T}} \sum_{k=0}^{j-1} |c_{t-k-1}| \left[E \left(\left(\frac{x_{t-k-1}}{\sqrt{T}} \right)^2 \right) \right]^{1/2}; \end{aligned}$$

and therefore using the uniform boundedness of the variance of $T^{-1/2}x_{t-k-1}$, it follows indeed that $\left[E \left(|x_t - x_{t-j}|^2 \right) \right]^{1/2} \leq Cj^{1/2}$ as required.

We then follow [Breitung and Demetrescu \(2015, Proof of Theorem 2\)](#) and obtain via partial summation that,

$$\begin{aligned} \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} x_{t-1} &= \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T (S_{t-1} - S_{t-2}) x_{t-1} \\ &= \frac{1}{T^{1+\eta}} (S_{T-1} x_{T-1} - S_{p-1} x_p) - \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} \Delta x_{t-1}. \end{aligned}$$

Now, since $S_{p-1} x_p = O_p(1)$ it is negligible in the limit; furthermore note that,

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} \Delta x_{t-1} = \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} v_{t-1} - \frac{1}{T^{2+\eta}} \sum_{t=p+1}^T c_{t-2} S_{t-2} x_{t-2}.$$

For the first summand on the r.h.s., we have using the Phillips–Solo device for the AR process v_{t-1} that,

$$\begin{aligned} \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} v_{t-1} &= \frac{\omega}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} v_{t-1} + \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} \Delta \tilde{v}_{t-1} \\ &=: A_T + B_T, \end{aligned}$$

where $\tilde{v}_t$ is a linear process with exponentially decaying coefficients.

Then, $A_T = \frac{\omega}{aT} \sum_{t=p+1}^T x_{t-2} v_{t-1} + \frac{\omega}{T^{1/2}} \sum_{t=p+1}^T R_{t-2,T} v_{t-1}$. It is furthermore seen from the expression of $R_{t,T}$ ([Breitung and Demetrescu, 2015, Proof of Theorem 2](#)) that $R_{t,T}$ is independent of $u_{t+j}, v_{t+j} \forall 1 \leq j \leq T-t$ whenever $(u_t, v_t)'$ is serially independent, such that $R_{t-2,T} v_{t-1}$ are the elements of a martingale difference array with uniformly vanishing variance, so $\text{Var} \left(\sum_{t=p+1}^T R_{t-2,T} v_{t-1} \right) = o_p(T)$ as required for the summand involving $R_{t,T}$ to vanish. Since v_{t-1} is independent of x_{t-2} and the conditions of [Hansen \(1992\)](#) are fulfilled, $T^{-1} \sum_{t=p+1}^T x_{t-2} v_{t-1}$ converges weakly, and we obtain

$$A_T \Rightarrow \frac{\omega^2}{a} \int_0^1 X(s) dM(s).$$

Using the partial summation formula on B_T , it follows that,

$$B_T = \frac{1}{T^{1+\eta}} (\tilde{v}_{T-1} S_{T-2} - \tilde{v}_{p-1} S_{p-1}) - \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T \tilde{v}_{t-2} \Delta S_{t-2}. \quad (\text{A.9})$$

Since $\sup_{1 \leq t \leq T} |S_t| = T^\eta \sup_{1 \leq t \leq T} |x_t| + o_p(T^{1/2+\eta}) = O_p(T^{1/2+\eta})$ and $\tilde{v}_{p-1} S_{p-1} = O_p(1)$, it follows that the first summand on the r.h.s. of (A.9); for the second, we have

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T \tilde{v}_{t-2} \Delta S_{t-2} = \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T \tilde{v}_{t-2} z_{t-2}.$$

Clearly, $\tilde{v}_{t-2}$ is uniformly L_2 -bounded, and it is easily shown that $T^{-\eta/2} z_t$ is uniformly L_2 -bounded as well. Then, the Cauchy–Schwarz inequality indicates that $E(|\tilde{v}_{t-2} z_{t-2}|) < CT^{\eta/2}$ such that

$$E \left| \frac{1}{T^{1+\eta}} \sum_{t=p+1}^T \tilde{v}_{t-2} \Delta S_{t-2} \right| \leq CT^{-\eta/2}$$

and $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T \tilde{v}_{t-2} \Delta S_{t-2}$ vanishes in probability.
Hence

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} x_{t-1} = \frac{1}{a} \frac{x_{T-1}^2}{T} - \frac{1}{a} \left(\frac{a\omega}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} v_{t-1} - \frac{1}{T^2} \sum_{t=p+1}^T c_{t-2} x_{t-2}^2 \right) + o_p(1).$$

Using the weak convergence of S_t and x_t we obtain

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} x_{t-1} \Rightarrow \frac{\omega^2}{a} \left[X^2(1) - \left(\int_0^1 X(s) dM(s) - \int_0^1 c(s) X^2(s) ds \right) \right],$$

and also

$$\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} \bar{x} \Rightarrow \frac{\omega^2}{a} X(1) \int_0^1 X(s) ds$$

as required. Note that, interestingly, $\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T S_{t-2} v_{t-1}$ converges under serial correlation to an Itô-type integral without bias term, unlike $\frac{1}{T} \sum_{t=p+1}^T x_{t-2} v_{t-1}$. This is because S_t and x_t require different normalizations, and the B_T term defined above vanishes for $\eta > 0$ – but not for $\eta = 0$, which is essentially characterizing $\frac{1}{T} \sum_{t=p+1}^T x_{t-2} v_{t-1}$. ■

Proof of Theorem 3.1. Write for the stable autoregression case $\hat{v}_t := \ddot{v}_t - (\hat{\phi} - \phi)' \ddot{x}_{t-p}$ with $\ddot{x}_{t-p}$ stacking the p lags of $\ddot{x}_t$ and ϕ the corresponding coefficients (of $(1 - \rho L) A(L)$), i.e. the pure autoregressive representation of x_t .

Note that $\tilde{y}_t = \ddot{y}_t - \hat{\gamma} \hat{v}_t$ since $\hat{v}_t$ is demeaned by construction, and $\ddot{y}_t = \beta \ddot{x}_{t-1} + \hat{\gamma} \hat{v}_t + \gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t + \ddot{\varepsilon}_t$ such that

$$\tilde{\beta}^{ivx} := \frac{\sum_{t=p+1}^T z_{t-1} \tilde{y}_t}{\sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1}} = \beta + \frac{\sum_{t=p+1}^T z_{t-1} (\gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t + \ddot{\varepsilon}_t)}{\sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1}}. \quad (\text{A.10})$$

Then, analyze

$$\begin{aligned} z_{t-1} &= \sum_{j=0}^{t-3} \varrho^j \Delta x_{t-1-j} \\ &= (x_{t-1} - \mu_x) - \varrho^{t-3} (x_1 - \mu_x) + (\varrho - 1) \sum_{j=0}^{t-4} \varrho^j (x_{t-2-j} - \mu_x). \end{aligned}$$

We have that $(\varrho - 1) \sum_{j=0}^{t-4} \varrho^j (x_{t-2-j} - \mu_x) = -\frac{a}{T^\eta} \sum_{j=0}^{t-4} \varrho^j (x_{t-2-j} - \mu_x) = -\frac{a}{T^\eta} d_{t-2}$ where d_{t-2} is here, with x_t a stable autoregression, a mildly integrated process which is known to be $O_p(T^{\eta/2})$. Furthermore, $\varrho^{t-3} \rightarrow 0$ when t goes to infinity at suitable rates; in the derivations below, the effect will be quantified precisely whenever needed, but it is important to keep in mind that $z_{t-1} \approx x_{t-1} - \mu_x$ which is a stable autoregression.

We thus have for the numerator of $\tilde{\beta}^{ivx} - \beta$ in (A.10) that,

$$\sum_{t=p+1}^T z_{t-1} (\ddot{\varepsilon}_t + \gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t) = \sum_{t=p+1}^T z_{t-1} \ddot{\varepsilon}_t - \gamma \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - \ddot{v}_t) - (\hat{\gamma} - \gamma) \sum_{t=p+1}^T z_{t-1} \hat{v}_t. \quad (\text{A.11})$$

The first two summands in (A.11) deliver a normal distribution. This is because $\sum_{t=p+1}^T z_{t-1} \ddot{\varepsilon}_t = \sum_{t=p+1}^T z_{t-1} \varepsilon_t + o_p(T^{1/2})$ and

$$\begin{aligned} \frac{1}{T^{1/2}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t &= \frac{1}{T^{1/2}} \sum_{t=p+1}^T x_{t-1} \varepsilon_t - \frac{a}{T^{1/2+\eta}} \sum_{t=p+1}^T d_{t-2} \varepsilon_t + \frac{x_1}{T^{1/2}} \sum_{t=p+1}^T \varrho^{t-3} \varepsilon_t \\ &= \frac{1}{T^{1/2}} \sum_{t=p+1}^T x_{t-1} \varepsilon_t + o_p(1) \end{aligned}$$

with $\sum_{t=p+1}^T d_{t-2} \varepsilon_t = O_p(T^{1/2+\eta/2})$ given the results in the proofs of Lemmas A.2 and A.3, and $\sum_{t=p+1}^T \varrho^{t-3} \varepsilon_t = O_p(T^{\eta/2})$ given that $\text{Var}\left(\sum_{t=p+1}^T \varrho^{t-3} \varepsilon_t\right) = O_p\left(\sum_{t=p+1}^T \varrho^{2t}\right) = O_p(T^\eta)$. Furthermore, since $\sum_{t=p+1}^T z_{t-1} \ddot{v}_t = \sum_{t=p+1}^T z_{t-1} v_t + o_p(T^{1/2})$,

$$\frac{1}{T^{1/2}} \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - \ddot{v}_t) = -\left(\frac{1}{T} \sum_{t=p+1}^T z_{t-1} \ddot{x}'_{t-p}\right) \sqrt{T} (\hat{\mathbf{a}} - \mathbf{a}) + o_p(1),$$

where the OLS autoregressive estimators, $\sqrt{T} (\hat{\mathbf{a}} - \mathbf{a}) = \left(\frac{1}{T} \sum_{t=p+1}^T \ddot{x}_{t-p} \ddot{x}'_{t-p}\right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=p+1}^T \ddot{x}_{t-p} v_t$, following standard arguments can be shown to have a limiting multivariate normal distribution. We now show that $\frac{1}{T} \sum_{t=2}^T z_{t-1} \ddot{x}_{t-p}$ does not converge to a vector of zeros, such that the limiting distribution of $\frac{1}{T^{1/2}} \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - v_t)$ is driven by $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T (\mathbf{x}_{t-p} - \mu_x \iota) v_t$ with ι a vector of ones (noting that demeaning x_t has no effect asymptotically). Given that

Table B.1

Empirical rejection frequencies of **right-sided tests** against local alternatives (positive short-run AR parameter ($a_1 = 0.5$) and error correlation of -0.95).

b		t_{IUX}	$\tilde{t}_{\text{IUX}}^{\mu_0}$	$\tilde{t}_{\text{IUX}}^{\mu_1}$	$\tilde{t}_{\text{IUX}}^{\mu_2}$	$\tilde{t}_{\text{IUX}}^{\mu_3}$	t_{IUX}	$\tilde{t}_{\text{IUX}}^{\mu_0}$	$\tilde{t}_{\text{IUX}}^{\mu_1}$	$\tilde{t}_{\text{IUX}}^{\mu_2}$	$\tilde{t}_{\text{IUX}}^{\mu_3}$
		T = 200					T = 500				
c = 0	0	0.119	0.058	0.148	0.050	0.070	0.118	0.060	0.149	0.050	0.063
	5	0.995	0.815	0.973	0.897	0.837	0.994	0.843	0.978	0.914	0.850
	10	1.000	0.971	0.999	0.991	0.974	1.000	0.980	0.998	0.993	0.980
	15	1.000	0.993	1.000	0.999	0.994	1.000	0.997	1.000	0.999	0.997
	25	1.000	0.999	1.000	1.000	0.999	1.000	1.000	1.000	1.000	1.000
c = 10	0	0.098	0.056	0.123	0.047	0.066	0.102	0.059	0.123	0.050	0.062
	5	0.816	0.702	0.888	0.585	0.761	0.838	0.731	0.910	0.610	0.757
	10	1.000	0.994	1.000	1.000	0.997	1.000	0.996	1.000	1.000	0.997
	15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 20	0	0.086	0.056	0.100	0.043	0.065	0.088	0.055	0.105	0.043	0.061
	5	0.508	0.427	0.568	0.323	0.468	0.530	0.449	0.597	0.336	0.468
	10	0.990	0.970	0.996	0.940	0.984	0.995	0.984	0.998	0.961	0.988
	15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 30	0	0.079	0.055	0.092	0.040	0.063	0.082	0.055	0.094	0.040	0.059
	5	0.373	0.311	0.418	0.232	0.342	0.397	0.334	0.444	0.244	0.349
	10	0.885	0.841	0.913	0.737	0.868	0.913	0.880	0.939	0.784	0.894
	15	1.000	0.998	1.000	0.996	1.000	1.000	0.999	1.000	0.998	1.000
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 40	0	0.077	0.055	0.088	0.040	0.064	0.079	0.058	0.090	0.039	0.061
	5	0.303	0.251	0.336	0.186	0.276	0.324	0.273	0.359	0.196	0.284
	10	0.746	0.700	0.788	0.574	0.731	0.790	0.748	0.831	0.620	0.765
	15	0.987	0.978	0.993	0.941	0.985	0.994	0.991	0.997	0.969	0.993
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 50	0	0.074	0.057	0.084	0.039	0.064	0.076	0.057	0.087	0.040	0.061
	5	0.261	0.214	0.289	0.159	0.234	0.277	0.234	0.309	0.168	0.245
	10	0.638	0.586	0.678	0.463	0.618	0.682	0.633	0.722	0.508	0.649
	15	0.941	0.916	0.953	0.842	0.934	0.966	0.955	0.976	0.897	0.962
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Note: See Table 3.

demeaning x_t is easily seen to have no asymptotic effect in

$$\frac{1}{T} \sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-p} = \frac{1}{T} \sum_{t=p+1}^T (x_{t-1} - \mu_x) \ddot{x}_{t-p} - \frac{1}{T} \sum_{t=p+1}^T \varrho^{t-3} (x_1 - \mu_x) \ddot{x}_{t-p} - \frac{a}{T^{1+\eta}} \sum_{t=p+1}^T d_{t-2} \ddot{x}_{t-p},$$

the first summand on the r.h.s. gives the desired limit (see Lemma A.1). The second is easily seen to vanish since $E((x_1 - \mu_x)(x_t - \mu_x))$ vanishes at exponential rate (in t). For the third, we show that $\sum_{t=p+1}^T d_{t-2} \ddot{x}_{t-p} = O_p(T)$ as follows. First, note that demeaning x_t has no asymptotic effect, then, by resorting to the Phillips–Solo device, it is tedious, yet straightforward to show that

$$\frac{1}{T} \sum_{t=p+1}^T d_{t-2} (\mathbf{x}_{t-p} - \mu_x) = O_p \left(\frac{1}{T} \sum_{t=p+1}^T \tilde{d}_{t-2} v_{t-p} \right) \quad \text{where} \quad \tilde{d}_{t-2} := \sum_{j=0}^{t-3} \varrho^j v_{t-2-j}.$$

Then, $\frac{1}{T} \sum_{t=p+1}^T \tilde{d}_{t-2} v_{t-p} = \frac{1}{T} \sum_{t=p+1}^T \tilde{d}_{t-p-1} v_{t-p} + O_p(1)$, and the proofs of Lemmas A.2 and A.3 provide the arguments leading to $\frac{1}{T} \sum_{t=p+2}^T \tilde{d}_{t-p-1} v_{t-p} = O_p \left(\frac{T^{1/2+\eta/2}}{T} \right) = O_p(1)$ as required.

The third summand in (A.11) is

$$\frac{\hat{\gamma} - \gamma}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} \hat{v}_t = (\hat{\gamma} - \gamma) \left(\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} v_t + \frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - v_t) \right) = o_p(1)$$

Table B.2

Empirical rejection frequencies of **two-sided tests** against local alternatives (positive short-run AR parameter ($a_1 = 0.5$) and error correlation of -0.95).

	b	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
c = 0	0	0.057	0.030	0.074	0.027	0.035	0.057	0.031	0.077	0.030	0.034
	5	0.976	0.733	0.936	0.818	0.759	0.975	0.767	0.946	0.841	0.775
	10	1.000	0.948	0.994	0.978	0.954	0.999	0.963	0.995	0.985	0.965
	15	1.000	0.986	0.999	0.996	0.988	1.000	0.992	0.999	0.998	0.992
	25	1.000	0.998	1.000	1.000	0.998	1.000	0.999	1.000	1.000	0.999
c = 10	0	0.056	0.035	0.064	0.064	0.040	0.061	0.036	0.067	0.064	0.037
	5	0.651	0.526	0.745	0.407	0.585	0.674	0.554	0.773	0.427	0.584
	10	1.000	0.984	1.000	0.998	0.992	1.000	0.989	1.000	1.000	0.992
	15	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 20	0	0.058	0.043	0.062	0.073	0.045	0.058	0.041	0.062	0.070	0.042
	5	0.354	0.278	0.407	0.208	0.315	0.371	0.296	0.426	0.220	0.312
	10	0.966	0.923	0.981	0.851	0.949	0.979	0.948	0.990	0.897	0.961
	15	1.000	0.999	1.000	1.000	0.999	1.000	1.000	1.000	1.000	1.000
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 30	0	0.057	0.047	0.060	0.075	0.048	0.057	0.042	0.060	0.071	0.045
	5	0.249	0.194	0.285	0.144	0.218	0.262	0.211	0.300	0.152	0.223
	10	0.773	0.713	0.820	0.592	0.753	0.819	0.764	0.860	0.640	0.783
	15	0.998	0.991	0.999	0.979	0.997	1.000	0.998	1.000	0.992	0.999
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 40	0	0.056	0.049	0.060	0.076	0.050	0.057	0.045	0.059	0.071	0.046
	5	0.193	0.156	0.221	0.113	0.173	0.209	0.169	0.236	0.123	0.177
	10	0.607	0.545	0.651	0.423	0.583	0.654	0.592	0.700	0.470	0.610
	15	0.959	0.936	0.973	0.866	0.953	0.980	0.969	0.986	0.921	0.974
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
c = 50	0	0.057	0.050	0.061	0.077	0.051	0.056	0.046	0.057	0.069	0.046
	5	0.162	0.135	0.182	0.100	0.149	0.177	0.146	0.198	0.103	0.153
	10	0.482	0.430	0.529	0.324	0.461	0.533	0.484	0.575	0.367	0.497
	15	0.868	0.830	0.891	0.732	0.853	0.916	0.892	0.935	0.799	0.903
	25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Note: See Table 3.

since $\hat{\gamma}$ is easily shown to be consistent for γ , $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} v_t = O_p(1)$ like in the case of $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t$, and $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - v_t) = O_p(1)$ as above. Hence,

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} (\varepsilon_t + \gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t) &= \frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t \\ &+ \gamma \left(\frac{1}{T} \sum_{t=p+1}^T z_{t-1} (\mathbf{x}'_{t-p} - \mu_x \mathbf{l}') \right) \left(\frac{1}{T} \sum_{t=p+1}^T (\mathbf{x}_{t-p} - \mu_x \mathbf{l}) (\mathbf{x}'_{t-p} - \mu_x \mathbf{l}') \right)^{-1} \times \\ &\times \frac{1}{\sqrt{T}} \sum_{t=p+1}^T (\mathbf{x}_{t-p} - \mu_x \mathbf{l}) v_t + o_p(1). \end{aligned}$$

Furthermore, it is shown along the lines of the discussion of $T^{-1} \sum_{t=p+1}^T z_{t-1} (\mathbf{x}_{t-p} - \mu_x \mathbf{l})$ that

$$\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t = \frac{1}{\sqrt{T}} \sum_{t=p+1}^T (\mathbf{x}_{t-1} - \mu_x) \varepsilon_t + o_p(1).$$

for both $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t$ and $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T (\mathbf{x}_{t-p} - \mu_x \mathbf{l}) v_t$, Theorem 24.3 in Davidson (1994) is easily checked to apply (see Lemma A.1 for the convergence of the sample covariance matrices); since $(\mathbf{x}_{t-p} - \mu_x \mathbf{l}) v_t$ and $z_{t-1} \varepsilon_t$ are orthogonal thanks to the uncorrelatedness of v_t and ε_t , the term $\frac{1}{\sqrt{T}} \sum_{t=p+1}^T z_{t-1} (\varepsilon_t + \gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t)$ is asymptotically normal with mean zero

Table B.3Empirical rejection frequencies of **right-sided tests** under breaks in variance and persistence, positive short-run AR parameter ($a_1 = 0.5$) and error correlation of -0.95 .

$c(s)$	$\sigma_\varepsilon^2 = \sigma_v^2$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$	t_{ivx}	$\tilde{t}_{ivx}^{\mu_0}$	$\tilde{t}_{ivx}^{\mu_1}$	$\tilde{t}_{ivx}^{\mu_2}$	$\tilde{t}_{ivx}^{\mu_3}$
		T = 200					T = 500				
-5	1	0.117	0.060	0.142	0.056	0.070	0.116	0.064	0.145	0.059	0.069
	1 + 8I ($s > 0.3$)	0.145	0.067	0.158	0.064	0.078	0.145	0.065	0.161	0.064	0.070
	1 + 8I ($s > 0.7$)	0.210	0.075	0.221	0.078	0.087	0.203	0.070	0.220	0.070	0.075
	9 - 8I ($s > 0.3$)	0.182	0.076	0.241	0.058	0.087	0.183	0.068	0.245	0.054	0.071
	9 - 8I ($s > 0.7$)	0.142	0.065	0.182	0.060	0.076	0.134	0.062	0.175	0.054	0.067
5 + 5I ($s > 0.5$)	1	0.111	0.061	0.148	0.057	0.073	0.110	0.063	0.147	0.058	0.069
	1 + 8I ($s > 0.3$)	0.136	0.067	0.153	0.068	0.077	0.136	0.063	0.154	0.068	0.067
	1 + 8I ($s > 0.7$)	0.189	0.077	0.209	0.077	0.087	0.183	0.069	0.205	0.072	0.074
	9 - 8I ($s > 0.3$)	0.188	0.083	0.251	0.062	0.095	0.184	0.076	0.250	0.057	0.080
	9 - 8I ($s > 0.7$)	0.139	0.069	0.186	0.060	0.081	0.130	0.067	0.180	0.055	0.071
5 + 20I ($s > 0.5$)	1	0.102	0.063	0.143	0.056	0.076	0.100	0.063	0.142	0.052	0.069
	1 + 8I ($s > 0.3$)	0.121	0.067	0.146	0.068	0.077	0.121	0.060	0.145	0.067	0.064
	1 + 8I ($s > 0.7$)	0.152	0.071	0.175	0.067	0.081	0.146	0.063	0.173	0.064	0.068
	9 - 8I ($s > 0.3$)	0.193	0.094	0.254	0.059	0.106	0.186	0.084	0.254	0.055	0.088
	9 - 8I ($s > 0.7$)	0.133	0.075	0.182	0.054	0.086	0.124	0.069	0.177	0.049	0.073
20	1	0.088	0.060	0.104	0.044	0.070	0.081	0.056	0.094	0.040	0.061
	1 + 8I ($s > 0.3$)	0.120	0.064	0.128	0.061	0.072	0.114	0.063	0.124	0.062	0.067
	1 + 8I ($s > 0.7$)	0.185	0.082	0.194	0.074	0.089	0.181	0.073	0.192	0.067	0.076
	9 - 8I ($s > 0.3$)	0.157	0.076	0.187	0.043	0.082	0.164	0.075	0.192	0.040	0.078
	9 - 8I ($s > 0.7$)	0.113	0.067	0.137	0.047	0.076	0.104	0.062	0.127	0.043	0.067
25 - 5I ($s > 0.5$)	1	0.090	0.058	0.105	0.045	0.069	0.084	0.057	0.098	0.042	0.061
	1 + 8I ($s > 0.3$)	0.123	0.064	0.132	0.062	0.074	0.118	0.063	0.128	0.061	0.067
	1 + 8I ($s > 0.7$)	0.194	0.083	0.204	0.076	0.093	0.190	0.074	0.200	0.070	0.077
	9 - 8I ($s > 0.3$)	0.153	0.073	0.183	0.042	0.080	0.160	0.072	0.189	0.040	0.075
	9 - 8I ($s > 0.7$)	0.113	0.066	0.138	0.049	0.074	0.106	0.062	0.128	0.043	0.066
25 - 20I ($s > 0.5$)	1	0.103	0.056	0.108	0.049	0.069	0.105	0.060	0.109	0.049	0.064
	1 + 8I ($s > 0.3$)	0.141	0.062	0.145	0.060	0.075	0.141	0.067	0.144	0.064	0.070
	1 + 8I ($s > 0.7$)	0.233	0.086	0.236	0.082	0.098	0.226	0.079	0.230	0.075	0.083
	9 - 8I ($s > 0.3$)	0.134	0.059	0.153	0.036	0.067	0.138	0.056	0.153	0.034	0.059
	9 - 8I ($s > 0.7$)	0.118	0.055	0.126	0.046	0.066	0.113	0.052	0.116	0.041	0.056

Note: See Table 5.

and asymptotic variance

$$\alpha_0 \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds + \gamma^2 (\alpha_0 \dots \alpha_{p-1}) \Omega^{-1} (\alpha_0 \dots \alpha_{p-1})' \int_0^1 \sigma_v^4(s) ds.$$

Checking that $\frac{1}{T} \sum_{t=p+1}^T z_{t-1}^2 \hat{\varepsilon}_t^2 + \frac{1}{T} \hat{\gamma}^2 \hat{Q}_T$ estimates the above asymptotic variance consistently is straightforward and we omit the details. Extending the derivations to alternative mean adjustment schemes in the autoregression delivering the residuals $\hat{v}_t$ is standard in the (locally) stationary case and we omit the details. ■

Proof of Theorem 3.2. Standard OLS algebra shows that the residuals $\hat{v}_t$ are numerically the same as in the autoregressive representation of x_t if resorting to the error-correction representation, which is more convenient with near-integration. We may thus write $\hat{v}_t := \tilde{v}_t - (\hat{\phi} - \phi) \ddot{x}_{t-1} - (\hat{\alpha} - \alpha)' \Delta \ddot{x}_{t-p+1}$ with $\Delta \ddot{x}_{t-p+1}$ stacking the first $p-1$ lags of $\Delta x_t - \bar{\Delta x}$ and $\phi := \frac{1}{\omega}(\rho - 1)$ (the vector α depends on all autoregressive coefficients of x_t , but its exact value is irrelevant here).

We have the same representation as in (A.11), but, since $\sum_{t=p+1}^T z_{t-1} = O_p(T^{1/2+\eta})$ (see Lemma A.4 item 3), we have that $\sum_{t=p+1}^T z_{t-1} \bar{\varepsilon} = \sum_{t=p+1}^T z_{t-1} \bar{v} = O_p(T^\eta)$ such that, with $\eta < 1$,

$$\sum_{t=p+1}^T z_{t-1} (\ddot{\varepsilon}_t + \gamma \ddot{v}_t - \hat{\gamma} \hat{v}_t) = \sum_{t=p+1}^T z_{t-1} \varepsilon_t - \gamma \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - v_t) - (\hat{\gamma} - \gamma) \sum_{t=p+1}^T z_{t-1} \hat{v}_t + o_p(T^{1/2+\eta/2}),$$

Table B.4

Empirical rejection frequencies of **two-sided tests** under breaks in variance and persistence, positive short-run AR parameter ($a_1 = 0.5$) and error correlation of -0.95 .

$c(s)$	$\sigma_\varepsilon^2 = \sigma_v^2$	t_{IVX}	$\tilde{t}_{IVX}^{\mu_0}$	$\tilde{t}_{IVX}^{\mu_1}$	$\tilde{t}_{IVX}^{\mu_2}$	$\tilde{t}_{IVX}^{\mu_3}$	t_{IVX}	$\tilde{t}_{IVX}^{\mu_0}$	$\tilde{t}_{IVX}^{\mu_1}$	$\tilde{t}_{IVX}^{\mu_2}$	$\tilde{t}_{IVX}^{\mu_3}$
		T = 200					T = 500				
-5	1	0.061	0.032	0.075	0.047	0.038	0.066	0.035	0.080	0.054	0.038
	1 + 8I ($s > 0.3$)	0.092	0.035	0.094	0.040	0.042	0.091	0.037	0.098	0.039	0.040
	1 + 8I ($s > 0.7$)	0.160	0.050	0.166	0.053	0.056	0.152	0.040	0.159	0.044	0.044
	9 - 8I ($s > 0.3$)	0.125	0.045	0.158	0.115	0.049	0.126	0.037	0.159	0.125	0.039
	9 - 8I ($s > 0.7$)	0.081	0.034	0.104	0.055	0.039	0.078	0.032	0.096	0.060	0.033
5 + 5I ($s > 0.5$)	1	0.063	0.034	0.078	0.060	0.039	0.066	0.037	0.082	0.066	0.040
	1 + 8I ($s > 0.3$)	0.095	0.040	0.097	0.048	0.045	0.093	0.037	0.096	0.044	0.038
	1 + 8I ($s > 0.7$)	0.152	0.051	0.159	0.054	0.055	0.141	0.041	0.152	0.045	0.043
	9 - 8I ($s > 0.3$)	0.132	0.050	0.172	0.129	0.054	0.134	0.041	0.171	0.137	0.043
	9 - 8I ($s > 0.7$)	0.081	0.037	0.112	0.066	0.043	0.081	0.036	0.107	0.072	0.037
5 + 20I ($s > 0.5$)	1	0.063	0.037	0.081	0.073	0.043	0.059	0.041	0.078	0.077	0.042
	1 + 8I ($s > 0.3$)	0.092	0.044	0.098	0.055	0.048	0.086	0.041	0.094	0.052	0.041
	1 + 8I ($s > 0.7$)	0.128	0.052	0.140	0.056	0.054	0.125	0.046	0.135	0.052	0.047
	9 - 8I ($s > 0.3$)	0.142	0.055	0.180	0.140	0.061	0.140	0.047	0.177	0.144	0.049
	9 - 8I ($s > 0.7$)	0.084	0.043	0.113	0.072	0.048	0.086	0.042	0.107	0.083	0.043
20	1	0.057	0.043	0.062	0.066	0.045	0.057	0.046	0.062	0.072	0.047
	1 + 8I ($s > 0.3$)	0.093	0.046	0.091	0.052	0.049	0.088	0.045	0.090	0.048	0.046
	1 + 8I ($s > 0.7$)	0.167	0.057	0.168	0.058	0.062	0.162	0.050	0.163	0.050	0.051
	9 - 8I ($s > 0.3$)	0.146	0.061	0.163	0.147	0.063	0.158	0.054	0.170	0.150	0.054
	9 - 8I ($s > 0.7$)	0.086	0.048	0.096	0.081	0.051	0.086	0.048	0.092	0.086	0.049
25 - 5I ($s > 0.5$)	1	0.058	0.042	0.063	0.061	0.045	0.059	0.044	0.061	0.067	0.045
	1 + 8I ($s > 0.3$)	0.093	0.045	0.091	0.050	0.048	0.089	0.044	0.089	0.047	0.045
	1 + 8I ($s > 0.7$)	0.175	0.058	0.172	0.058	0.062	0.166	0.049	0.165	0.047	0.050
	9 - 8I ($s > 0.3$)	0.140	0.059	0.157	0.142	0.061	0.153	0.053	0.163	0.147	0.052
	9 - 8I ($s > 0.7$)	0.085	0.047	0.093	0.076	0.050	0.084	0.047	0.090	0.082	0.047
25 - 20I ($s > 0.5$)	1	0.059	0.035	0.061	0.044	0.039	0.061	0.041	0.063	0.048	0.042
	1 + 8I ($s > 0.3$)	0.092	0.039	0.087	0.040	0.043	0.092	0.042	0.092	0.041	0.044
	1 + 8I ($s > 0.7$)	0.188	0.058	0.187	0.056	0.063	0.178	0.047	0.177	0.044	0.049
	9 - 8I ($s > 0.3$)	0.115	0.052	0.122	0.123	0.051	0.117	0.049	0.121	0.132	0.048
	9 - 8I ($s > 0.7$)	0.071	0.040	0.075	0.057	0.041	0.070	0.044	0.071	0.062	0.043

Note: See Table 5.

yet z_t is now a mildly integrated variable. Still, Lemmas A.3 and A.4 show that $\frac{1}{T^{1/2+\eta/2}} \sum_{t=p+1}^T z_{t-1} \varepsilon_t$ is asymptotically normal with variance $\omega^2 \int_0^1 \sigma_v^2(s) \sigma_\varepsilon^2(s) ds$, whereas the remaining term can be re-written as

$$\frac{1}{T^{1/2+\eta/2}} \sum_{t=p+1}^T z_{t-1} (\hat{v}_t - v_t) = -\frac{1}{T^{1/2+\eta/2}} \sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1} (\hat{\phi} - \phi) - \frac{1}{T^{1/2+\eta/2}} \sum_{t=p+1}^T z_{t-1} \Delta \ddot{x}'_{t-p+1} (\hat{\alpha} - \alpha).$$

In the limit, this vanishes because $(\hat{\phi} - \phi)$ is $O_p(T^{-1})$ and $(\hat{\alpha} - \alpha) = O_p(T^{-1/2})$ as standard analysis of near-unit root autoregressions shows, while, at the same time, $\sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1} = O_p(T^{1+\eta})$ (see Lemma A.4.3) and we only need to show that $\sum_{t=p+1}^T z_{t-1} \Delta \ddot{x}'_{t-p+1} = O_p(T)$. This is known to be the case when z_{t-1} is a near-integrated or stationary variable; we discuss here the case where z_t is an IVX instrument. Examining $\sum_{t=p+2}^T z_{t-1} \Delta \ddot{x}_{t-1}$ as a representative for the whole vector,

$$\frac{1}{T} \sum_{t=p+1}^T z_{t-1} \Delta \ddot{x}_{t-1} = \frac{1}{T} \sum_{t=p+1}^T z_{t-1} v_{t-1} + \frac{1}{T^2} \sum_{t=p+1}^T c_t z_{t-1} x_{t-2} - \frac{1}{T} \sum_{t=p+1}^T z_{t-1} \overline{\Delta x},$$

it is easily shown that both $\frac{z_t}{\sqrt{T}}$ and $\frac{x_t}{\sqrt{T}}$ are uniformly L_2 -bounded, hence $E\left(\frac{1}{T^2} \sum_{t=p+1}^T c_t z_{t-1} x_{t-2}\right) = O(1)$. Moreover, $\frac{1}{T} \sum_{t=p+1}^T z_{t-1} v_{t-1}$ is itself $O_p(1)$, which can be shown along the lines of the discussion for $\frac{1}{T} \sum_{t=2}^T q_{t-2} x_{t-p}$ in the proof of Theorem 3.1, while $\sum_{t=p+1}^T z_{t-1} \overline{\Delta x} = O_p(T^\eta)$ like above. This completes the result for residuals from an OLS-demeaned autoregression.

Should one demean by subtracting the initial value, i.e. work with $\ddot{x}_t^* = x_t - x_1$ in the autoregression (but not in the predictive regression), we note that $\Delta \ddot{x}_t^* = \Delta \ddot{x}_t$, and also that the weak convergence of the levels x_t is not affected;

neither are the convergence rates of the coefficients of the autoregression. The result then follows immediately. Should one consider recursive demeaning in the autoregression, we note that only the levels require this in the error-correction representation, which then does not affect the convergence rates of the estimators (cf. Shin and So, 2001), as required. ■

Proof of Theorem 3.3. Since the residual effect of ε_t and v_t is easily checked to be negligible on the standard error estimator, the correction Q_T is negligible under the local alternative as well and we have for the residual-augmented IVX t -statistic that,

$$\begin{aligned}\tilde{t}_{\beta_1}^{ivx} &= \frac{\sum_{t=p+1}^T z_{t-1} (\varepsilon_t + \beta_1 \ddot{x}_{t-1})}{\sqrt{\sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2}} + o_p(1) \\ &= \frac{\sum_{t=p+1}^T z_{t-1} \ddot{\varepsilon}_t}{\sqrt{\sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2}} + b \frac{\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1} \ddot{x}_{t-1}}{\sqrt{\frac{1}{T^{1+\eta}} \sum_{t=p+1}^T z_{t-1}^2 \varepsilon_t^2}} + o_p(1).\end{aligned}$$

The first summand on the r.h.s. converges to a standard normal distribution, $\mathcal{Z}$; note that $\mathcal{Z}$ would indeed be independent of the limit process of the regressor x_t since $z_{t-1}\varepsilon_t$ and v_t are orthogonal. Thus, the result follows with Lemma A.4, items 1 and 3 upon noting that demeaning ε_t has no asymptotic effect. ■

Appendix B. Additional tables

See Tables B.1–B.4.

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